

# PMATH 367

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Fall 2024

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# 1 Topological Spaces

## Motivation:

1.  $A \in \mathbb{R}^n$  closed  $\iff R^n \setminus A$  open
2.  $x_n \rightarrow x \in \mathbb{R}^n \iff \forall \text{ open } x \in U, \exists x \in \mathbb{N} \text{ such that } n \geq N \implies x_n \in U$
3.  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is cts  $\iff f^{-1}(U)$  is open in  $R^n$  whenever  $U$  is open in  $\mathbb{R}^m$
4.  $A \subseteq R^n$  is compact  $\iff$  every open cover of  $A$  has a finite subcover

Big ideal: All of these concepts from analysis can be stated using open sets.

Recall:

$P(X) = \{A : A \subseteq X\}$  is the power set of  $X$ .

**Definition 1.1.** Let  $X$  be a set. We say  $\mathcal{T} \subseteq P(X)$  is a topology on  $X$  iff

1.  $\emptyset, X \in \mathcal{T}$
2. If  $I$  is an index set and  $A_\alpha \in \mathcal{T}$  for all  $\alpha \in I$ , then  $\bigcup_{\alpha \in I} A_\alpha \in \mathcal{T}$
3. finite intersection of  $A \in \mathcal{T}$  is still in  $\mathcal{T}$

We call  $(X, \mathcal{T})$  a topological space. We call the elements of  $\mathcal{T}$  the open sets of  $X$ . The closed subsets of  $X$  are exactly  $X \setminus A, A \in \mathcal{T}$

**Big Idea:** Topology, the study of topological spaces, is the area of math which studies concepts like opens/closed sets, continuity, compactness, and things like connectedness.

**Example 1.2.**  $X = \{a, b, c\}, \mathcal{T}_1 = \{\emptyset, \{a, b\}, \{c\}, X\}, \mathcal{T}_2 = \{\emptyset, \{a, b\}, \{b, c\}, \{b\}, X\}$

**Example 1.3.**  $(X, d)$  metric space, let  $\mathcal{T} = \{U \subseteq X : \forall x \in U, \exists r > 0, B_r(x) \subseteq U\}$  is the metric topology on  $X$ .

**Example 1.4.** Why is  $\mathcal{T}_1$  above is not a metric topology?

If  $d$  was a metric on  $X = \{a, b, c\}$ , then  $\exists r_1, r_2, r_3 > 0$  such that  $B_{r_1}(a) = \{a\}, B_{r_2}(b) = \{b\}, B_{r_3}(c) = \{c\}$ . Thus, the metric topology would be  $P(X)$

**Example 1.5.**  $X$  any set, then  $P(X)$  discrete topology,  $\{\emptyset, X\}$  indiscrete topology.

**Example 1.6.**  $X$  any set,  $\mathcal{T}_f = \{A \subseteq X : X \setminus A \text{ finite}\} \cup \{\emptyset\}$  finite-complement topology.

Why?  $X \setminus X = \emptyset$  is finite hence  $X \in \mathcal{T}_f, \emptyset \in \mathcal{T}_f$  by design.

$A_\alpha \in \mathcal{T}_f \implies X \setminus A_\alpha$  finite, hence  $X \setminus (\bigcup A_\alpha) = \bigcap (X \setminus A_\alpha)$  finite by de morgan's law.

$X \setminus A, X \setminus A$  finite,  $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$  is finite.

**Example 1.7.**  $X$  set,  $\mathcal{T}_c = \{A \subseteq X : X \setminus A \text{ is at most countable}\} \cup \{\emptyset\}$  is the countable complement topology.

**Definition 1.8.**  $X$  set. We say  $\mathcal{B} \subseteq P(X)$  is a basis for a topology on  $X$  iff

1.  $\forall x \in X, \exists B \in \mathcal{B}, x \in B$

2.  $\forall x \in X$  such that  $x \in B_1 \cap B_2$ , where  $B_1, B_2 \in \mathcal{B}$ , there exists  $B_3 \in \mathcal{B}$  such that  $x \in B_3 \subseteq B_1 \cap B_2$

**Example 1.9.**  $X = \mathbb{R}$ ,  $\mathcal{B} = \{(a, b) : a < b\}$  is a basis for a topology on  $\mathbb{R}$ .

**Example 1.10.**  $(X, d)$ ,  $\mathcal{B} = \{B_r(x) : x \in X, r > 0\}$  is a basis for a topology on  $X$ .

**Example 1.11.**  $X$  set.  $\mathcal{B} = \{\{x\} : x \in X\}$  is a basis for a topology.

**Definition 1.12.** Let  $\mathcal{B}$  be a basis for a topology on  $X$ . We define the topology generated by  $\mathcal{B}$  to be

$$\mathcal{T}_{\mathcal{B}} = \{U \subseteq X : \forall x \in U, \exists B \in \mathcal{B}, x \in B \subseteq U\}$$

We say  $\mathcal{B}$  is a basis for  $\mathcal{T}_{\mathcal{B}}$

**Proposition 1.13.** If  $\mathcal{B}$  is a basis, then  $\mathcal{T}_{\mathcal{B}}$  is a topology.

*Proof.*  $\emptyset \in \mathcal{T}_{\mathcal{B}}$ . Also,  $\forall x \in X, \exists B \in \mathcal{B}, x \in B \subseteq X \implies X \in \mathcal{T}_{\mathcal{B}}$

$U_{\alpha} \in \mathcal{T}_{\mathcal{B}}, x \in \bigcup_{\alpha} U_{\alpha}$ , say  $x \in U_{\beta}$ , implies that  $\exists B \in \mathcal{B}$  such that  $x \in B \subseteq U_{\beta} \subseteq \bigcup_{\alpha} U_{\alpha} \implies \bigcup_{\alpha} U_{\alpha} \in \mathcal{T}_{\mathcal{B}}$ .

$U, V \in \mathcal{T}_{\mathcal{B}}$ . Take  $x \in U \cap V$ ,  $\exists B_1, B_2 \in \mathcal{B}$  such that

$$x \in B_1 \subseteq U, x \in B_2 \subseteq V$$

$$\implies \exists B_3 \in \mathcal{B} \text{ such that } x \in B_3 \subseteq B_1 \cap B_2 \subseteq U \cap V$$

$$\implies U \cap V \subseteq \mathcal{T}_{\mathcal{B}}$$

□

**Remark 1.14.**  $\forall B \in \mathcal{B}, B \in \mathcal{T}_{\mathcal{B}}$  by taking  $U = B$  in this case.

**Proposition 1.15.** Let  $\mathcal{B}$  be a basis for a topology on  $X$ . Then

$$\mathcal{T}_{\mathcal{B}} = \{\bigcup_{\alpha} B_{\alpha} : \alpha \in I, B_{\alpha} \in \mathcal{B}\}$$

*Proof.* Let  $\mathcal{U}$  denote the set on the right. Let  $U \in \mathcal{T}_{\mathcal{B}}$ . For all  $x \in U$ ,  $\exists B_x \in \mathcal{B}$  such that  $x \in B_x \subseteq U$ . Thus,  $U = \bigcup_{x \in U} B_x$ . Since each  $B_{\alpha} \in \mathcal{T}_{\mathcal{B}}$ , and  $\mathcal{T}_{\mathcal{B}}$  is a topology,  $\mathcal{U} \subseteq \mathcal{T}_{\mathcal{B}}$  □

**Example 1.16.**  $X = \mathbb{R}$ ,  $\mathcal{B} = \{(a, b) : a < b\}$ . Then  $\mathcal{T}_{\mathcal{B}} =$  metric/standard topology.

**Example 1.17.**  $(X, d)$ .  $\mathcal{B} = \{B_r(x)\}$ . Then  $\mathcal{T}_{\mathcal{B}} =$  metric topology.

**Example 1.18.**  $X$ ,  $\mathcal{B} = \{\{x\} : x \in X\}$ . Then  $\mathcal{T}_{\mathcal{B}} = P(X)$  discrete topology.

**Example 1.19.**  $\mathcal{B}' = \{[a, b) : a < b\}$  is a basis for topology on  $\mathbb{R}$ . Let  $\mathcal{T}' = \mathcal{T}_{\mathcal{B}'}$ , and let  $\mathcal{T}$  be a metric topology on  $\mathbb{R}$ .

Claim:  $\mathcal{T} \not\subseteq \mathcal{T}'$

$[0, 1) \in \mathcal{T}' \setminus \mathcal{T}$ , so they are not the same.  $(a, b) = \bigcup_{n \in \mathbb{N}} [a + \frac{1}{n}, b) \in \mathcal{T}'$ .

We call  $\mathcal{T}'$  the lower limit topology.

**Definition 1.20.**  $X$  set. We say  $S \subseteq P(X)$  is a subbasis for a topology on  $X$  if and only if

$$X = \bigcup_{A \in S} A$$

The topology generated by  $S$  is

$$\mathcal{T}_S = \{\bigcup_{\alpha} (A_{\alpha_1} \cap \dots \cap A_{\alpha_n}) : n \in \mathbb{N}, A_{\alpha_i} \in S\}$$

**Proposition 1.21.** Let  $S$  be a subbasis for a topology on  $X$ , then

$\mathcal{B} = \{A_1 \cap \dots \cap A_n : n \in \mathbb{N}, A_i \in S\}$  is a basis for a topology on  $X$ . Thus, we have  $\mathcal{T}_S = \mathcal{T}_{\mathcal{B}}$

*Proof.*  $X = \bigcup_{A \in S} A$ . Then  $x \in X \implies x \in A$  for some  $A \in S$ . Since  $A \in \mathcal{B}$ , this proves the first axiom of a basis. Say

$$x \in (A_1 \cap \dots \cap A_n) \cap (B_1 \cap \dots \cap B_m) \in \mathcal{B}$$

□

**Definition 1.22.**  $(X, \mathcal{T}), A \subseteq X$ . Then  $\mathcal{T}_A = \{A \cap U : U \in \mathcal{T}\}$  is a topology on  $A$ . We call it the subspace topology on  $A$ . We say  $A$  is a subspace of  $X$ .

**HW:**  $(X, \mathcal{T}), A \subseteq X$ . If  $\mathcal{B}$  is a basis for  $\mathcal{T}$ , then  $\mathcal{B}' = \{A \cap B : B \in \mathcal{B}\}$  is a basis for  $\mathcal{T}_A$ .

**Example 1.23.**  $A = [0, 2] \subseteq \mathbb{R}$ . Then  $(1, 3) \cap A = (1, 2]$  open in  $A$  but not in  $\mathbb{R}$ .

**Proposition 1.24.**  $(X, \mathcal{T}), U \in \mathcal{T}$ . If  $A \subseteq U$  is open in  $U$ , then  $A$  is open in  $X$ .

**Proposition 1.25.**  $(X, \mathcal{T}), A \subseteq X$ . Then the closed subsets of  $A$  are exactly the sets of the form  $A \cap C$ , where  $C$  is closed in  $X$ .

*Proof.* Suppose  $C \subseteq A$  is closed, then  $A \setminus C$  is open in  $A$ . Hence  $A \setminus C = A \cap U, U \in \mathcal{T}$ , which implies  $C = A \setminus (A \cap U) = A \cap (X \setminus U)$ , where  $X \setminus U$  is closed in  $X$ . Moreover, if  $C$  is closed in  $X$ , then  $X \setminus C \in \mathcal{T}$ , and hence  $A \setminus (A \cap C) = A \cap (X \setminus C) \in \mathcal{T}_A$ . □

**Example 1.26.**  $A = [0, 1] \cup (2, 3) \subseteq \mathbb{R}$ . Then  $[0, 1] = A \cap (-1, 2)$  open in  $A$ , and  $[0, 1] = A \cap [0, 1]$  closed in  $A$ . Hence  $[0, 1]$  is clopen in  $A$ .

## 2 Closed Sets

**Remark 2.1.** For  $(X, \mathcal{T})$ :

1.  $\emptyset, X$  are closed
2. closed sets are "closed" under arbitrary intersections
3. Finite unions of closed sets are closed.

**Definition 2.2.**  $(X, \mathcal{T}), A \subseteq X$ :

1. The closure of  $A$  is

$$\overline{A} = \cap \{C \subseteq X : A \subseteq C, C \text{ closed in } X\}$$

the smallest closed set containing  $A$

2. the interior of  $A$  is

$$\text{int}(A) = \cup \{U \in \mathcal{T} : U \subseteq A\}$$

largest open set contained in  $A$

**Remark 2.3.**  $\text{int}(A) \subseteq A \subseteq \overline{A}$

**Definition 2.4.**  $(X, \mathcal{T})$ . If  $x \in X$  and  $U \in \mathcal{T}$  with  $x \in U$ , we say  $U$  is a neighborhood (nbhd) of  $x$ .

**Proposition 2.5.**  $(X, \mathcal{T}), A \subseteq X$ . Then  $x \in \overline{A}$  iff for all nbhd's  $U$  of  $x$ ,  $U \cap A \neq \emptyset$ .

*Proof.* Let  $x \in \overline{A}$ . For contradiction, suppose  $\exists U \in \mathcal{T}$  such that  $x \in U$  and  $U \cap A = \emptyset$ . Then  $A \subseteq X \setminus U$ , and hence  $\overline{A} \subseteq X \setminus U$ , which means  $x \in U$  and  $x \in \overline{A} \subseteq X \setminus U$ , contradiction.

Suppose  $x \in X$  such that  $\forall x \in U \in \mathcal{T}, U \cap A \neq \emptyset$ . Let  $C \subseteq X$  be closed such that  $A \subseteq C$ . Note that  $X \setminus C$  is open. Suppose for contradiction,  $x \notin C$ , then  $x \in X \setminus C$ , and hence  $(X \setminus C) \cap A \neq \emptyset$ , contradiction. Thus,  $x \in C$  and  $x \in \overline{A}$ .  $\square$

**Definition 2.6.**  $(X, \mathcal{T}), A \subseteq X$ . We say  $x \in X$  is a limit point of  $A$  iff every nbhd of  $x$  intersects  $A$  at a point different from  $x$ .

**Example 2.7.**  $X = \mathbb{R}, A = (0, 1) \cup \{2\}$ , then  $0, \frac{1}{2}, 1$  are limit points of  $A$ ,  $2$  is not a limit point of  $A$ .

**Definition 2.8.**  $(X, \mathcal{T}), A \subseteq X$ , we write  $A' = \{x \in X : \text{limit point of } A\}$

**Proposition 2.9.**  $(X, \mathcal{T}), A \subseteq X, \overline{A} = A \cup A'$

*Proof.* ( $\Leftarrow$ ) Trivial

( $\Rightarrow$ ) Let  $x \in \overline{A}$ . Let  $x \in U \in \mathcal{T}$ . Thus,  $U \cap A \neq \emptyset$ .

If  $x \in U \cap A \Rightarrow x \in A$ . If  $x \notin U \cap A \Rightarrow x \in A'$ .  $\square$

**Corollary 2.10.**  $(X, \mathcal{T}), A \subseteq X$ .  $A$  is closed iff  $A' \subseteq A$ .

*Proof.*  $A' \subseteq A \iff A = \overline{A}$   $\square$

### 3 Hausdorff Space

**Definition 3.1.**  $(X, \mathcal{T})$ . We say  $X$  is Hausdorff iff  $\forall x \neq y \in X, \exists U, V \in \mathcal{T}$  such that

1.  $x \in U, y \in V$
2.  $U \cap V = \emptyset$

**Example 3.2.** Metric  $\implies$  Hausdorff

$x \neq y, \epsilon = d(x, y)$ , we have  $x \in B_{\epsilon/2}(x), y \in B_{\epsilon/2}(y)$

**Example 3.3.**  $X = \{a, b, c\}, \mathcal{T} = \{\emptyset, X, \{a, b\}, \{c\}\}$  is not Hausdorff ( $a \neq b$ )

**Example 3.4.**  $(\mathbb{R}, \mathcal{T}_f)$  is not Hausdorff:

$x \neq y, x \in U, y \in V$ , then

$$x \in U = \mathbb{R} \setminus \{x_1, \dots, x_n\}$$

$$y \in V = \mathbb{R} \setminus \{y_1, \dots, y_m\}$$

which implies  $U \cap V \neq \emptyset$

**Proposition 3.5.**  $(X, \mathcal{T})$  is Hausdorff. Then  $\{x\}$  is closed.

*Proof.*  $x \in X$ ,  $y \neq x$ , we can find  $x \in U_y \in \mathcal{T}$ ,  $y \in V_y \in \mathcal{T}$  and  $U_y \cap V_y = \emptyset$

$$X \setminus \{x\} = \bigcup_{y \neq x} V_y$$

which is open, and hence  $\{x\}$  is closed. □

**Remark 3.6.**  $(X, \mathcal{T})$  Hausdorff, then every finite subset of  $X$  is closed.

**Proposition 3.7.**  $(X, \mathcal{T})$  Hausdorff,  $A \subseteq X$ . Then  $x \in X$  is a limit point of  $A$  iff every nbhd of  $x$  intersects  $A$  at infinitely many points.

*Proof.* ( $\Leftarrow$ ) Trivial

( $\Rightarrow$ ) Assume  $x$  is a limit point of  $A$ . For contradiction, assume  $\exists x \in U \in \mathcal{T}$  such that  $U \cap A$  is finite. By assumption, say  $(U \cap (A \setminus \{x\})) = \{x_1, \dots, x_n\} \neq \emptyset$ . Then  $\{x_1, \dots, x_n\}$  is closed and hence  $V = X \setminus \{x_1, \dots, x_n\}$  is open, and

$$x \in U \cap V (\text{open})$$

However,

$$A \cap (U \cap V) = \{x\} \text{ OR } \emptyset$$

Either of these reaches a contradiction. □

## 4 Continuity

**Definition 4.1.**  $(X, \mathcal{T}), (Y, \mathcal{U})$ . We say  $f : x \rightarrow y$  is continuous (cts) iff

$$\forall U \in \mathcal{U}, f^{-1}(U) \in \mathcal{T}$$

where  $f^{-1}(U) = \{x \in X : f(x) \in U\}$

**Proposition 4.2.**  $(X, \mathcal{T}), (Y, \mathcal{U}), f : X \rightarrow Y$ . If  $\mathcal{B}$  is a basis for  $\mathcal{U}$ , then  $f$  is cts iff  $f^{-1}(B) \in \mathcal{T}$  for all  $B \in \mathcal{B}$ .

*Proof.* ( $\Leftarrow$ )  $f^{-1}(B) \in \mathcal{T}, U \in \mathcal{U}$ , then  $U = \cup B_i \implies f^{-1}(U) = \cup f^{-1}(B_i) \in \mathcal{T}$

( $\Rightarrow$ )  $f$  cts,  $B \in \mathcal{B} \implies B \in \mathcal{U}, f^{-1}(B) \in \mathcal{T}$  □

HW: The same results holds for a subbasis.

**Proposition 4.3.**  $(X, \mathcal{T}), (Y, \mathcal{U}), f : X \rightarrow Y$ , then TFAE:

1.  $f$  is cts
2.  $\forall A \subseteq X, f(\overline{A}) \subseteq \overline{f(A)}$
3.  $\forall$  closed  $C \subseteq Y, f^{-1}C$  is closed in  $X$

*Proof.* (1)  $\implies$  (2) :

Suppose  $f$  is cts. Let  $y = f(x) \in f(\overline{A})$  where  $x \in \overline{A}$ . Let  $U \in \mathcal{U}$  with  $y \in U$ . Then  $x \in f^{-1}(U) \in \mathcal{T}$  since  $f$  is cts. Since  $x \in \overline{A}$ , there exists  $a \in A$  such that  $a \in f^{-1}(U)$ . Hence  $f(a) \in U$  and  $f(a) \in f(A)$ . Thus,  $y \in \overline{f(A)}$ , which finishes the proof.

(2)  $\implies$  (3) :

Assume  $f(\overline{A}) \subseteq \overline{f(A)}$  for all  $A \subseteq X$ . Let  $C \subseteq Y$  be closed and let  $D = f^{-1}(C)$ . Let  $x \in \overline{D}$ . Then  $f(x) \in f(\overline{D}) \subseteq \overline{f(D)} \subseteq \overline{C} = C$ . Hence  $x \in f^{-1}(C) = D \implies \overline{D} \subseteq D \implies \overline{D} = D$ . Thus,  $D$  is closed.

(3)  $\implies$  (1) :

Let  $U \in \mathcal{U} \implies Y \setminus U$  closed. Hence  $f^{-1}(Y \setminus U) = X \setminus f^{-1}(U)$  closed, which gives  $f^{-1}(U)$  is open.  $\square$

**Example 4.4.**  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = \arctan(x)$  is cts.  $A = \mathbb{R}$ ,  $f(\overline{A}) = f(\mathbb{R}) = (-\frac{\pi}{2}, \frac{\pi}{2})$  while  $\overline{f(A)} = \overline{(-\frac{\pi}{2}, \frac{\pi}{2})} = [-\frac{\pi}{2}, \frac{\pi}{2}]$

## 5 Homeomorphisms

**Definition 5.1.**  $f : X \rightarrow Y$ , where  $(X, \mathcal{T}), (Y, \mathcal{U})$ . We say  $f$  is a homeomorphism (homeo) iff

1.  $f$  is bijective ( $f$  is invertible)
2.  $f$  cts
3.  $f^{-1}$  cts

**Example 5.2.**  $X = \mathbb{Z}, \mathcal{T} = P(\mathbb{N}) \cup \{\mathbb{Z}\}$ . Let  $f : X \rightarrow X, x \mapsto x - 1$  which is bijective.

Then  $A \subseteq \mathbb{N} \implies f^{-1}(A) \subseteq \mathbb{N}$  and  $f^{-1}(\mathbb{Z}) = \mathbb{Z}$ . Thus,  $f$  is cts.

Let  $g(x) = f^{-1}(x) = x + 1$ . Note that  $\{1\} \in \mathcal{T}$ , but  $g^{-1}(\{1\}) = \{0\} \notin \mathcal{T}$ , thus  $f^{-1}(x)$  is not cts.

**Example 5.3.**  $S^1 = \{(x, y) : x^2 + y^2 = 1\}$ . Let  $f[0, 1) \rightarrow S^1, x \mapsto (\cos(2\pi x), \sin(2\pi x))$ . So  $f$  is cts and  $f$  is bijective by construction. However,  $f^{-1}$  is not cts as

$$(f^{-1})^{-1}([0, \frac{1}{4})) = f([0, \frac{1}{4}))$$

where  $[0, \frac{1}{4})$  is open but  $f([0, \frac{1}{4}))$  is not (in subspace topology sense).

Big Idea:  $[0, 1)$  and  $S^1$  have topological/geometrical differences and so there cannot exist a homeo between them. For instance,  $S^1$  compact while  $[0, 1)$  is not.

**Definition 5.4.**  $f : X \rightarrow Y$ . We say  $f$  is an open map iff  $f(U) \subseteq Y$  is open whenever  $U \subseteq X$  is open.

**Remark 5.5.**  $f : X \rightarrow Y$  is a homeo iff

1.  $f$  is bijective
2.  $f$  cts
3.  $f$  open map

*Proof.*  $(f^{-1})^{-1}(U) = f(U)$  □

Big Idea: Suppose  $f : X \rightarrow Y$  is a homeo.

Then for the points: every  $y \in Y$  is of the form  $y = f(x)$  for a unique  $x \in X$ , and hence we can think  $Y$  is a relabelling of  $X$ .

For the open sets: the elements of  $\mathcal{U}$  are exactly  $f(U), U \in \mathcal{T}$ . If  $V \in \mathcal{U}$ , then  $f^{-1}(V) \in \mathcal{T}$  and  $V = f(f^{-1}(U))$ , and hence we can think  $\mathcal{U}$  is a relabelling of  $\mathcal{T}$ .

Thus,  $(X, \mathcal{T}), (Y, \mathcal{U})$  are the same topological spaces up to the relabelling  $f$ .

**Example 5.6.**  $f : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2}), x \mapsto \arctan(x)$  is a homeomorphism.

**Proposition 5.7.** If  $f : X \rightarrow Y$  is constant, then  $f$  is cts.

*Proof.* Say  $f(x) = y_0$ . If  $U \subseteq Y$  open, then  $f^{-1}(U) = X, y_0 \in U$  or  $\emptyset, y_0 \notin U$  □

**Proposition 5.8.**  $A \subseteq X$ . Then  $i : A \rightarrow X, i(x) = x$  is cts.

*Proof.* If  $U \subseteq X$  is open, then  $i^{-1}(U) = A \cap U$  is open. □

**Proposition 5.9.**  $f : X \rightarrow Y, g : Y \rightarrow Z$  cts. Then  $g \circ f : X \rightarrow Z$  is cts.

*Proof.* If  $U \subseteq Z$  is open, then

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$$

□

**Proposition 5.10.** If  $f : X \rightarrow Y$  is cts and  $A \subseteq X$ , then  $f|_A : A \rightarrow Y$  is cts.

*Proof.*  $(f|_A)^{-1}(U) = A \cap f^{-1}(U)$  is open if  $U \subseteq Y$  is open. □

HW: Prove the proposition below

**Proposition 5.11.**  $f : X \rightarrow Y$  is cts

1. if  $f(X) \subseteq Z \subseteq Y$ , then  $f : X \rightarrow Z$  is cts.
2. if  $Y \subseteq Z$ , then  $f : X \rightarrow Z$  is cts.

**Proposition 5.12.**  $X = \bigcup U_\alpha, U_\alpha$  is open.  $f : X \rightarrow Y$ . If  $f|_{U_\alpha}$  is cts for all  $\alpha$ , then  $f$  is cts.

*Proof.* If  $V \subseteq Y$  is open, then  $f^{-1}(V) = f^{-1}(V) \cap X = \bigcup (f^{-1}(V) \cap U_\alpha) = \bigcup ((f|_{U_\alpha})^{-1}(V))$  □

**Definition 5.13.** We say  $f : X \rightarrow Y$  is cts at  $x \in X$  iff  $\forall$  open  $V \subseteq Y, f(x) \in V$ , there exists open  $U \subseteq X$  such that  $f(U) \subseteq V$ .

**Proposition 5.14.**  $f : X \rightarrow Y$  is cts iff  $f$  is cts at all  $x \in X$ .

*Proof.* ( $\implies$ ) Suppose  $f$  is cts and fix  $x \in X$ . If  $V$  is a nbhd of  $f(x)$ , then  $f^{-1}(V)$  is a nbhd of  $x$ . Moreover,  $f(f^{-1}(V)) \subseteq V$ .

( $\impliedby$ ) Suppose  $f$  is cts at all  $x \in X$ . Let  $V \subseteq Y$  be open. Let  $x \in f^{-1}(V)$ . By assumption, exists  $U_x$  open such that  $x \in U_x$  and  $f(U_x) \subseteq V$ . Hence  $U_x \subseteq f^{-1}(V)$ . Thus,  $f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$

open. □

**Proposition 5.15** (Pasting Lemma).  $X = A \cup B$ ,  $A, B$  closed. If  $f : A \rightarrow Y$  and  $g : B \rightarrow Y$  are cts and agree on  $A \cap B$ , then the natural  $h : X \rightarrow Y$  is cts.

*Proof.* Let  $C \subseteq Y$  be closed. Then

$$h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$$

where the former is closed in  $A$  and the latter is closed in  $B$ , and the former is closed in  $X$  and the latter is closed in  $X$ , which finishes the proof.  $\square$

Goal: Make new topologies from old topologies

Assignment 2:

1. Let  $X$  be a set and let  $\mathcal{F} = \{f_\alpha : X \rightarrow Y : \alpha \in A\}$ . Say  $(Y_\alpha, \mathcal{T}_\alpha)$  are topological spaces. Then

$$\mathcal{B} = \{f_{\alpha_1}^{-1}(U_1) \cap \dots \cap f_{\alpha_n}^{-1}(U_n) : U_i \in \mathcal{T}_{\alpha_i}\}$$

is the basis for a topology on  $X$ .

2. Then,  $\mathcal{T} = \mathcal{T}_{\mathcal{B}}$  is called the initial topology on  $X$  induced by  $\mathcal{F}$ . It is the smallest topology on  $X$  that makes every  $f_\alpha$  cts.
3. In  $X$ ,  $x_i \rightarrow x$  iff  $\forall \alpha, f_\alpha(x_i) \rightarrow f_\alpha(x)$
4.  $g : Z \rightarrow X$  is cts iff  $\forall \alpha, f_\alpha \circ g : Z \rightarrow Y_\alpha$  is cts.

## 6 Product Topology and Box Topology

### 6.1 Product Topology

Let  $(X_\alpha, \mathcal{T}_\alpha), \alpha \in A$  be a collection of topological spaces. Consider

$$X = \prod_{\alpha \in A} X_\alpha = \{f : A \rightarrow \cup X_\alpha \text{ s.t. } f(\alpha) \in X_\alpha\}$$

The product topology on  $X$  is the initial topology generated by

$$\mathcal{F} = \{\pi_\alpha : X \rightarrow X_\alpha \mid \alpha \in A\}$$

where  $\pi_\alpha(f) = f(\alpha)$ . We call  $\pi_\alpha$  the  $\alpha^{\text{th}}$  projection.

i.e., the product topology is the smallest topology on  $X$  which makes the projections cts.

Simple Case:  $(X, \mathcal{T}), (Y, \mathcal{U})$

$$\begin{aligned} X \times Y &= \{f : \{1, 2\} \rightarrow X \cup Y : f(1) \in X, f(2) \in Y\} \\ &\equiv \{(x, y) : x \in X, y \in Y\} \end{aligned}$$

We have

$$\pi_1(x, y) = x, \pi_2(x, y) = y$$

## 6.2 Box Topology

Let  $(X_\alpha, \mathcal{T}_\alpha), \alpha \in A$

HW:  $\mathcal{B} = \{\prod_\alpha U_\alpha : U_\alpha \in \mathcal{T}_\alpha\}$  is a basis for a topology on  $X = \prod X_\alpha$

Investigation:

Q: How do they differ?

Bases:

$$\mathcal{B}_p = \{\pi_{\alpha_1}^{-1}(U_1) \cap \dots \cap \pi_{\alpha_n}^{-1}(U_n)\}$$

$$\mathcal{B}_b = \{\prod_\alpha U_\alpha\}$$

Note:

$$\begin{aligned} & \pi_{\alpha_1}^{-1}(U_1) \cap \dots \cap \pi_{\alpha_n}^{-1}(U_n) \\ &= \{x \in \prod X_\alpha : \pi_{\alpha_1}(x) \in U_1, \dots, \pi_{\alpha_n}(x) \in U_n\} \\ &= \prod_\alpha V_\alpha, V_\alpha = \begin{cases} U_{\alpha_i}, & \alpha = \alpha_i \\ X_\alpha, & \alpha \neq \alpha_i \end{cases} \end{aligned}$$

Conclusion:

1.  $\mathcal{B}_p = \{\prod U_\alpha : \text{all but finitely many } U_\alpha = X_\alpha\}$
2.  $\mathcal{B}_p \subseteq \mathcal{B}_b \implies \text{prod} \subseteq \text{box}$
3. if  $A$  is finite, then  $\text{prod} = \text{box}$

**Example 6.1** (Warning).  $X = \prod_{i=1}^\infty \mathbb{R}$ ,  $f : \mathbb{R} \rightarrow X, f(t) = (t, t, t, \dots)$ .

$\forall n, \pi_n \circ f : \mathbb{R} \rightarrow \mathbb{R}, \pi_n(f(t)) = t$ . This is cts. By A2,  $f$  is cts wrt product topology. If

$$B = (-1, 1) \times (-\frac{1}{2}, \frac{1}{2}) \times \dots \in \text{Box}$$

Then  $f^{-1}(B) = 0$  not open, hence  $f$  is not cts wrt box.

## 7 Quotient Topology:

**Definition 7.1.** Let  $X$  be a set and let  $\sim$  be an equivalence relation on  $X$ . i.e.,

1.  $\forall x, x \sim x$
2.  $\forall x, y, x \sim y \implies y \sim x$
3.  $\forall x, y, z, (x \sim y \ \& \ y \sim z \implies x \sim z)$

For  $x \in X$ , let

$$[x] = \{y \in X : x \sim y\}$$

be the equivalence class containing  $x$ .

$$X/\sim := \{[x] : x \in X\}$$

**Example 7.2.**  $X = \mathbb{Z}, x \sim y \iff x \equiv y \pmod{n}$ , then  $X/\sim = \mathbb{Z}_n$

**Example 7.3.**  $X = [0, 1], x \sim y \iff \begin{cases} x = y \\ \text{OR} \\ x, y \in \{0, 1\} \end{cases}$

**Proposition 7.4.**  $(X, \mathcal{T}), X/\sim$ . Consider the quotient map:

$$q : X \rightarrow X/\sim, q(x) = [x]$$

Then  $\mathcal{Q} = \{U \subseteq X/\sim : q^{-1}(U) \subseteq \mathcal{T}\}$  is a topology on  $X/\sim$ , called the quotient topology on  $X/\sim$ .

Big Idea:  $\mathcal{Q}$  is the largest topology on  $X/\sim$  such that  $q$  is cts.

Why?

$q^{-1}(\emptyset) = \emptyset, q^{-1}(X/\sim) = X$ .  $U_\alpha \in \mathcal{Q}$ , then

$$q^{-1}(\cup U_\alpha) = \cup q^{-1}(U_\alpha) \in \mathcal{T}$$

$U, V \in \mathcal{Q}$ , then

$$q^{-1}(U \cap V) = q^{-1}(U) \cap q^{-1}(V) \in \mathcal{T}$$

**Proposition 7.5.**  $(X, \mathcal{T}), (Y, \mathcal{U})$ , a function  $f : X/\sim \rightarrow Y$  is cts iff  $f \circ q$  is cts.

*Proof.* ( $\implies$ ) Suppose  $f : X/\sim \rightarrow Y$  is cts. Since the composition of cts function is cts, we get the desired result.

( $\impliedby$ ) Suppose  $f \circ q$  is cts. For  $U \in \mathcal{U}$ ,  $(f \circ q)^{-1}(U) = q^{-1}(f^{-1}(U)) \in \mathcal{T}$ , hence  $f^{-1}(U) \in \mathcal{Q}$ .  $\square$

**Theorem 7.6** (Universal Property of Quotients). Let  $(X, \mathcal{T}), (Y, \mathcal{U})$  be topological spaces and let  $\sim$  be an equiv relation on  $X$ .

For every cts  $f : X \rightarrow Y$ , which is constant on equiv classes, there exists a unique (cts)  $\bar{f} : X/\sim \rightarrow Y$  such that

$$f = \bar{f} \circ q$$

*Proof.* Consider  $\bar{f} : X/\sim \rightarrow Y$  given by  $\bar{f}([x]) = f(x)$ . We have that

$$f(x) = \bar{f}([x]) = \bar{f}(q(x))$$

hence  $f = \bar{f} \circ q$ , by previous proposition,  $\bar{f}$  is cts.

If  $g$  is another such function, then  $\forall x \in X$ ,

$$g([x]) = g(q(x)) = f(x) = \bar{f}([x])$$

$\square$

**Example 7.7.**  $X = [0, 1], x \sim y \iff \begin{cases} x = y \\ x, y \in \{0, 1\} \end{cases}$  Goal: We want to show  $X/\sim \cong S^1$

Consider  $f : [0, 1] \rightarrow S^1$ ,  $f(x) = (\cos(2\pi x), \sin(2\pi x))$  is cts, surjective, and  $f(0) = f(1)$  (constant on equiv class). By UPQ, exists cts  $\bar{f} : X/\sim \rightarrow S^1$ ,  $f = \bar{f} \circ q$ .

Surjectivity:  $y \in S^1$ , exists  $x \in X$  such that  $f(x) = y$ , hence  $\bar{f}([x]) = y$

Injectivity:  $\bar{f}([x]) = \bar{f}([y]) \implies f(x) = f(y) \implies x \sim y \implies [x] = [y]$

Consider  $f^{-1}$ : Since  $[0, 1]$  is compact and  $q$  is cts,  $q(X) = X/\sim$  is compact. Moreover, since  $\bar{f}: X/\sim \rightarrow S^1$  is invertible, cts,  $X/\sim$  is compact, and  $S^1$  is Hausdorff,  $\bar{f}$  is a homeomorphism.

Culture:

In topology, we rarely give such proofs. We accept proofs by picture/gluing.

**Example 7.8** (Intuition Building).  $X = [0, 1] \times [0, 1]$

**Example 7.9.**  $X = [0, 1] \times [0, 1]$ ,  $X/\sim$  is mobius strip

**Example 7.10.**  $X = [0, 1] \times [0, 1]$ ,  $X/\sim$  is Klein Bottle

**Example 7.11.**  $\mathbb{D} = \{(x, y) : x^2 + y^2 \leq 1\}$

## 8 Connectedness

### 8.1 Connected Spaces

**Definition 8.1.**  $(X, \mathcal{T})$ ,

1. we say  $X = U \cup V$  is a separation of  $X$  iff

$$U, V \in \mathcal{T}, U, V \neq \emptyset, U \cap V = \emptyset$$

2. If a separation exists, we say  $X$  is separated
3. Otherwise, we say  $X$  is connected

**Example 8.2.**  $\mathbb{Q}$  with subspace topology on  $\mathbb{R}$ :

$$\mathbb{Q} = (-\infty, \pi) \cup (\pi, \infty)$$

**Example 8.3.**  $(\mathbb{R}, \mathcal{T}_f)$

Every two non-empty open sets intersect. Thus it is connected

**Proposition 8.4.**  $(X, \mathcal{T})$ . Then  $X$  is connected iff the only clopen subsets of  $X$  are  $\emptyset$  and  $X$ .

*Proof.* ( $\implies$ ): prove by contrapositive. Let  $A \subseteq X$  be clopen,  $A \neq \emptyset, A \neq X$ . Then

$$X = A \cup (X \setminus A)$$

is a separation. Thus,  $X$  is not connected.

( $\impliedby$ ):  $X = U \cup V$  be a separation.  $U \in \mathcal{T}, X \setminus U \in \mathcal{T}$ , hence  $U$  is open and closed, that is,  $U$  is clopen. However, the only clopen sets are  $\emptyset$  and  $X$ , hence one of  $U, V$  must be  $\emptyset$ , contradict to  $X = U \cup V$  is a separation.  $\square$

**Lemma 8.5.**  $X = U \cup V$  be a separation. If  $Y \subseteq X$  (subspace topology) is connected, then  $Y \subseteq U$  or  $Y \subseteq V$

*Proof.* First,  $Y = (Y \cap U) \cup (Y \cap V)$ , where they are open and disjoint. Since  $Y$  is connected,  $Y \cap U = \emptyset$  or  $Y \cap U = Y$ , hence  $Y \subseteq V$  or  $Y \subseteq U$ .  $\square$

**Proposition 8.6.**  $(X, \mathcal{T}), A_\alpha \in X$  connected. If  $\cap A_\alpha \neq \emptyset$ , then  $\cup A_\alpha$  is connected

*Proof.* Let  $Y = \cup A_\alpha$  and suppose  $Y = U \cup V$  is a separation. WLOG, say  $p \in U$ , where  $p \in \cap A_\alpha$ . By lemma,  $\forall \alpha, A_\alpha \subseteq U$ , hence  $Y \subseteq U$  and  $V = \emptyset$ , contradiction.  $\square$

**Example 8.7.**  $A_n = (n, n + \frac{1}{2})$  connected,  $\cup A_n$  separated.

**Proposition 8.8.**  $(X, \mathcal{T}), A \subseteq X$  connected. If  $A \subseteq B \subseteq \overline{A}$ , then  $B$  is connected. In particular,  $\overline{A}$  is connected.

*Proof.* Suppose, for contradiction,  $B = U \cup V$  is a separation of  $B$ . Then  $A \subseteq B$  and  $A$  is connected. WLOG, say  $A \subseteq U$ . Thus,  $\overline{A} \subseteq \overline{U}$ .

Claim:  $\overline{U} \cap V = \emptyset$

If  $x \in \overline{U} \cap V$ , then  $V$  is an open nbhd of  $x$ , hence  $U \cap V \neq \emptyset$ , contradiction

$B \cap V = \emptyset \implies V = \emptyset$ , contradiction.  $\square$

**Proposition 8.9.**  $X$  connected,  $f : X \rightarrow Y$  cts. Then  $f(X)$  is connected.

*Proof.* Suppose  $f(X) = U \cup V$  is a separation, then

$$X = \underbrace{f^{-1}(U) \cup f^{-1}(V)}_{\text{open, disjoint non-empty}}$$

, contradiction  $\square$

**Definition 8.10.**  $(X, \mathcal{T})$ . Define  $x \sim y$  iff  $\exists C \subseteq X$  connected s.t.  $x, y \in C$ . Then  $\sim$  is an equivalence relation.

The equivalence classes are called (connected) components of  $X$ .

**Remark 8.11.**  $(X, \mathcal{T})$

1. The components of  $X$  are pair-wise disjoint and partition  $X$ .
2. If  $A \subseteq X$  is connected, then  $A \subseteq C$  for a unique component  $C$

*Proof.* If  $x, y \in A$ , then  $x \sim y$ . In particular,  $x$  and  $y$  belong to the same equivalence class.  $\square$

**Proposition 8.12.** The connected components of  $X$  are connected.

*Proof.* Let  $C$  be a connected component of  $X$  and fix  $x_0 \in C$ . For  $x \in C$ ,  $x \sim x_0$ , and so there exists a connected  $x_0, x \in A_x \subseteq X$ . By the Remark,  $A_x \subseteq C$ , hence  $C = \bigcup_{x \in C} A_x$ . Since  $x_0 \in A_{x_0}$ ,  $C$  is connected.  $\square$

## 8.2 Path Connectedness

**Definition 8.13.**  $(X, \mathcal{T})$

1. A path from  $a \in X$  to  $b \in X$  is a cts

$$f : [0, 1] \rightarrow X$$

such that  $f(0) = a, f(1) = b$

2. We say  $X$  is path connected iff  $\forall a, b \in X$ , exists a path from  $a$  to  $b$  in  $X$ .

**Proposition 8.14.** Path Connected  $\implies$  Connected

*Proof.* Suppose  $X$  is path connected, but not connected:  $X = U \cup V$  is a separation of  $X$ . Take  $a \in U$  and  $b \in V$  and a path  $f : [0, 1] \rightarrow X$  from  $a$  to  $b$ . Then

$$[0, 1] = \underbrace{f^{-1}(U)}_{0 \in} \cup \underbrace{f^{-1}(V)}_{1 \in}, \text{ contradiction}$$

□

**Example 8.15** (Topologist's Sine Curve). Consider  $X = \underbrace{\{(x, \sin(\frac{1}{x})) : 0 < x < 1\}}_A \cup \{(0, 0)\}$ . Then

$A$  is path connected, and hence is connected. Thus,  $\overline{A} = X$  is connected. Suppose  $X$  is path connected, and let  $f$  be a path with  $f(0) = (0, 0), f(1) = (\frac{1}{\pi}, 0)$ . Write  $f(t) = (a(t), b(t))$ . Then IVT gives exists  $0 < t_1 < 1, a(t_1) = \frac{2}{3\pi}$ , and also gives exists  $0 < t_2 < 1, a(t_2) = \frac{2}{5\pi}$ . Continuing, there exists a decreasing  $(t_n) \subseteq [0, 1]$  such that  $a(t_n) = \frac{2}{(2n+1)\pi}$ . Then MCT:  $t_n \rightarrow t \in [0, 1] \implies b(t_n) \rightarrow b(t)$ , but  $\sin(\frac{(2n+1)\pi}{2}) = (-1)^n$

## 9 Compactness

**Definition 9.1.**  $(X, \mathcal{T})$

1. An open cover of  $X$  is a collection  $U_\alpha \in \mathcal{T} (\alpha \in A)$  such that  $X = \bigcup_{\alpha \in A} U_\alpha$ .
2. If  $B \subseteq A$  and

$$X = \bigcup_{\alpha \in B} U_\alpha$$

we call the  $U_\alpha$ 's ( $\alpha \in B$ ) a subcover. If  $|B| < \infty$ , we call it a finite subcover.

**Definition 9.2.**  $(X, \mathcal{T})$ . We say  $X$  is compact iff every open cover of  $X$  has a finite subcover.

**Big Idea:** Compactness is a bridge to finiteness "smallness"

**Example 9.3.**  $(X, \mathcal{T}_f)$ ,  $X = \bigcup_{\substack{\alpha \\ \neq \emptyset}} U_\alpha$  open cover. Fix  $U_0 = X \setminus \{x_1, x_2, \dots, x_n\}$ . Say  $x_i \in U_i$ , then  $X = U_0 \cup U_1 \cap \dots \cap U_n$ , hence  $X$  is compact.

**Example 9.4.**  $(\mathbb{R}, \mathcal{T}_c)$ . Let  $U_n = \mathbb{R} \setminus \{n, n+1, n+2, \dots\}, n \in \mathbb{N}$ . Then  $\mathbb{R} = \bigcup U_n$  but it has no finite subcover. Thus, it is not compact.

**Lemma 9.5.**  $(X, \mathcal{T}), A \subseteq X$ . Then  $A$  is compact iff whenever  $A \subseteq \bigcup U_\alpha, U_\alpha \in \mathcal{T}$ , then  $\exists \alpha_1, \alpha_2, \dots, \alpha_n$  such that

$$A \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$$

*Proof.*  $(\implies)$   $A \subseteq \bigcup U_\alpha \implies A = \bigcup (A \cap U_\alpha) = (A \cap U_{\alpha_1}) \cup \dots \cup (A \cap U_{\alpha_n}) \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$

$(\impliedby)$   $A = \bigcup (A \cap U_\alpha) \implies A \subseteq \bigcup U_\alpha \implies A \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n} \implies A = \bigcup_{i=1}^n (A \cap U_{\alpha_i})$  □

**Proposition 9.6.**  $(X, \mathcal{T})$  compact. If  $C \subseteq X$  is closed, then  $C$  is compact.

*Proof.* Suppose  $C \subseteq \bigcup U_\alpha, U_\alpha \in \mathcal{T}$ . Thus,  $X = (X \setminus C) \cup (\bigcup U_\alpha)$ . Since  $X$  is compact,

$$X = (X \setminus C) \cup U_{\alpha_1} \cup \dots \cup U_{\alpha_n} \text{ for some } \alpha_i$$

Hence  $C \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$  □

**Example 9.7.**  $(\mathbb{R}, \mathcal{T}_f)$  compact.

HW: All subsets of  $\mathbb{R}$  are compact. Then  $\mathbb{N}$  is compact but not closed.

**Proposition 9.8.**  $(X, \mathcal{T})$  Hausdorff. If  $K \subseteq X$  is compact, then  $K$  is closed.

*Proof.* Let  $K \subseteq X$  be compact. We want to show that  $X \setminus K$  is open. Fix  $x_0 \in X \setminus K$ . For all  $x \in K$ , there exists  $U_x, V_x \in \mathcal{T}$  such that  $U_x \cap V_x = \emptyset, x_0 \in U_x, x \in V_x$ . Then  $K \subseteq \bigcup_{x \in K} V_x$ , and since  $K$  is compact,

$$K \subseteq V_{x_1} \cup \dots \cup V_{x_n} \text{ for some } x_i \in K$$

Consider  $x_0 \in \underbrace{U_{x_1} \cap \dots \cap U_{x_n}}_U \in \mathcal{T}$ . Notice that  $x_0 \in U \subseteq X \setminus K$ , hence  $\text{int}(X \setminus K) = X \setminus K$  □

**Proposition 9.9.**  $X$  Hausdorff,  $K \subseteq X$  compact. For all  $x \in X \setminus K$ , exists  $U, V \in \mathcal{T}$  such that  $x \in U, K \subseteq V$ , and  $U \cap V = \emptyset$

**Proposition 9.10.**  $(X, \mathcal{T})$  compact.  $f : X \rightarrow Y$  cts. Then  $f(X)$  is compact.

*Proof.* Suppose  $f(X) \subseteq \bigcup U_\alpha, U_\alpha \subseteq Y$  is open. Then  $X = \bigcup f^{-1}(U_\alpha)$ , hence

$$X = f^{-1}(U_{\alpha_1}) \cup \dots \cup f^{-1}(U_{\alpha_n}) \text{ for some } \alpha_1, \dots, \alpha_n$$

$$\implies f(X) \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$$

□

**Proposition 9.11.**  $(X, \mathcal{T})$  compact,  $(Y, \mathcal{U})$  Hausdorff. If  $f : X \rightarrow Y$  is cts and bijective, then  $f$  is a homeomorphism.

*Proof.* We want show that if  $C \subseteq X$  is closed, then  $(f^{-1})^{-1}(C) = f(C)$  is closed.

Since  $X$  is compact, and  $C \subseteq X$  is closed,  $C$  is compact. Since  $f$  is cts,  $f(C)$  is compact. However,  $Y$  is Hausdorff and so  $C$  is also closed. □

**Theorem 9.12** (Tychonoff's Theorem). If  $(X_\alpha, \mathcal{T}_\alpha)$  are compact, then  $\prod X_\alpha$  is compact (product topology).

**Fact:** Tychonoff is equivalent to the axiom of choice.

**Definition 9.13.** Let  $X$  be a set. We say  $\leq$  is a partial order on  $X$  iff

1.  $\forall x \in X, x \leq x$
2.  $\forall x, y \in X, \text{ if } x \leq y \text{ and } y \leq x, \text{ then } x = y$
3.  $\forall x, y, z \in X, \text{ if } x \leq y, \text{ and } y \leq z, \text{ then } x \leq z$

We call  $(X, \leq)$  a poset. We say  $A \subseteq X$  where  $X$  is a poset a chain iff  $\forall a, b \in A, a \leq b \text{ or } b \leq a$ . We say  $x \in X$  is maximal iff  $\forall y \in X, x \leq y \implies x = y$ .

Let  $A \subseteq X$  be a chain. An upper bound for  $A$  is any  $x \in X$  such that  $\forall a \in A, a \leq x$ .

**Theorem 9.14** (Zorn's Lemma). Let  $(X, \leq)$  be a non-empty poset. If every chain in  $X$  has an upper bound, then  $X$  has a maximal element.

**Definition 9.15.** Let  $\mathcal{C} \subseteq P(X)$ . We say  $\mathcal{C}$  has the **finite intersection property (FIP)** if for all  $F_1, \dots, F_n \in \mathcal{C}$ , we have  $F_1 \cap \dots \cap F_n \neq \emptyset$ .

**Proposition 9.16.**  $(X, \mathcal{T})$ . Then  $X$  is compact if and only if whenever  $\mathcal{C}$  is a family of closed sets in  $X$  having FIP, we have  $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$

*Proof.*  $(\implies)$ : Homework.

$(\impliedby)$ : Suppose  $X$  satisfies the condition on such families of closed sets. Consider:

$$X = \bigcup_{U_\alpha \in \mathcal{U}} \implies X \setminus U_\alpha = \bigcap_{U_\alpha \in \mathcal{U}} (X \setminus U_\alpha)$$

Therefore  $\emptyset = (X \setminus U_{\alpha_1}) \cap \dots \cap (X \setminus U_{\alpha_n})$ , hence:

$$X = U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$$

as desired. □

**Lemma 9.17.**  $(X, \mathcal{T})$ . Let  $\mathcal{C} \subseteq P(X)$  be a family of closed sets having the FIP. There exists  $C \subseteq \mathcal{F} \subseteq P(X)$  which is maximal with respect having the FIP.

*Proof.* Let  $Y = \{K \subseteq P(X) : K \subseteq \mathcal{C}, K \text{ has the FIT}\}$ , and order  $Y$  via  $\subseteq$ . Note that  $\mathcal{C} \in Y$  so  $Y \neq \emptyset$ . Let  $S \subseteq Y$  be a chain. Consider  $Z = \bigcup_{A \in S} A$ . Note that  $\mathcal{C} \subseteq Z$  since  $Z \subseteq A$  for all  $A \in S$ .

Claim:  $Z$  has the FIP.

*claim.* Let  $F_1, \dots, F_n \subseteq Z$ . Say each  $F_i \in A_i \in S$ . Since  $S$  is a chain, WLOG suppose  $A_i \subseteq A_1, \forall i \in \{1, \dots, n\}$ . Then  $F_1, \dots, F_n \subseteq A_1$ . Hence:

$$F_1 \cap \dots \cap F_n \neq \emptyset$$

since  $A_1 \in Y$  has the FIP. Therefore  $Z$  has the FIP. □

By the claim,  $Z$  is an upper bound for the chain  $S$ . Hence by Zorn's lemma,  $Y$  has a maximal element  $\mathcal{F}$  as desired.  $\square$

**Lemma 9.18.**  $(X, \mathcal{T})$ . Let  $\mathcal{C} \subseteq \mathcal{F} \subseteq P(X)$  be as before.

1.  $\mathcal{F}$  is closed under finite intersections
2. If  $A \subseteq X$  intersects every  $F \in \mathcal{F}$ , then  $A \in \mathcal{F}$ .

*Proof.* (1) : Let  $F_1, \dots, F_n \in \mathcal{F}$ , then  $\mathcal{F} \cap \{F_1 \cap \dots \cap F_n\}$  has the FIP. By the maximality of  $\mathcal{F}$ , we get  $\mathcal{F} \cup \{F_1 \cap \dots \cap F_n\} = \mathcal{F}$ , and hence  $F_1 \cap \dots \cap F_n \in \mathcal{F}$

(2) : Note that  $\mathcal{F} \cup \{A\}$  has the FIP, then  $A \in \mathcal{F}$   $\square$

*Tychonoff Theorem.* Let  $\mathcal{C}$  be a family of closed sets in  $X = \prod X_\alpha$  having the FIP. Consider  $\mathcal{C} \subseteq \mathcal{F}$  maximal with respect to FIP. Define:

$$\mathcal{A}$$

$\square$

**Claim:**  $p \in \bigcap_{F \in \mathcal{F}} \overline{F} (\subseteq \bigcap_{C \in \mathcal{C}} C)$

*claim.* Suppose that  $p \in U = \pi_{\alpha_1}^{-1}(U_{\alpha_1}) \cap \dots \cap \pi_{\alpha_n}^{-1}(U_{\alpha_n})$ , where  $U_{\alpha_i} \in \mathcal{T}_{\alpha_i}$ , and for  $i = 1, 2, \dots, n$ :

$$\pi_{\alpha_i}(p) = P_{\alpha_i} \in U_{\alpha_i}$$

For all  $A \in \mathcal{A}_{\alpha_i}$ ,  $P_{\alpha_i} \in U_{\alpha_i} \cap \overline{A}$ . Hence  $U_{\alpha_i} \cap A \neq \emptyset$ . Then for all  $F \in \mathcal{F}$ , there exists  $z \in U_{\alpha_i} \cap \pi_{\alpha_i}(F)$ . Say  $z = \pi_{\alpha_i}(f)$ ,  $f \in F$ . Therefore  $f \in \pi_{\alpha_i}^{-1}(U_{\alpha_i}) \cap F$

$$\implies \forall F \in \mathcal{F} : F \cap \pi_{\alpha_i}^{-1}(U_{\alpha_i}) \neq \emptyset$$

By a lemma,  $\pi_{\alpha_i}^{-1}(U_{\alpha_i}) \in \mathcal{F}$ . By the other lemma,  $U \in \mathcal{F}$ . Then, for all  $F \in \mathcal{F}$ ,  $U, F \in \mathcal{F} \implies U \cap F \neq \emptyset$  by FIP. Thus,  $p \in \bigcap_{F \in \mathcal{F}} \overline{F}$ .  $\square$

## 10 Countability & Separation

**Definition 10.1.**  $(X, \mathcal{T})$ ,  $x \in X$ . A basis at  $x \in X$  is a collection  $\mathcal{B}$  of nbhd's of  $x$  such that whenever  $x \in U \in \mathcal{T}$ , then exists  $B \in \mathcal{B}$  such that  $B \subseteq U$

**Definition 10.2.** We say  $(X, \mathcal{T})$  is first countable iff  $\forall x \in X$ , exists a countable basis of  $x$ .

**Example 10.3.**  $(X, d)$ ,  $x \in X$ .  $\mathcal{B}_x = \{B_q(x) : q \in \mathbb{Q}\}$

**Idea:**  $(X, \mathcal{T})$  is first countable iff  $X$  has a strong relationship with countability (old school way).

**Proposition 10.4.**  $(X, \mathcal{T})$  first countable,  $A \subseteq X$

1.  $x \in \overline{A} \iff \exists (a_n) \in A, a_n \rightarrow x$
2.  $f : X \rightarrow Y$  is cts iff  $x_n \rightarrow x \implies f(x_n) \rightarrow f(x)$

*Proof.* (1) : (  $\Leftarrow$  ) covered by A1

(  $\Rightarrow$  ) Let  $\mathcal{B} = \{B_n : n \in \mathbb{N}\}$  be a basis at  $x$ . Take

$$\begin{aligned} a_1 &\in B_1 \cap A, \\ a_2 &\in B_1 \cap B_2 \cap A, \\ a_3 &\in B_1 \cap B_2 \cap B_3 \cap A \\ &\vdots \end{aligned}$$

**Claim:**  $a_n \rightarrow x$

*claim.* Let  $U \in \mathcal{T}$  with  $x \in U$ . There exists  $N \in \mathbb{N}$  such that

$$B_N \subseteq U$$

For all  $n \geq N$ ,

$$a_n \in B_N \subseteq U.$$

□

(2) (  $\Rightarrow$  ) : Covered by A1

(  $\Leftarrow$  ) : Let  $A \subseteq X$ .

**Claim:**  $f(\overline{A}) \subseteq \overline{f(A)}$

*claim.* Let  $y \in f(\overline{A})$  so that  $y = f(x)$  with  $x \in \overline{A}$ . By (1), there exists  $(a_n) \subseteq A$  such that  $a_n \rightarrow x$ .

$$\Rightarrow \underbrace{f(a_n)}_{\in f(A)} \rightarrow f(x) \Rightarrow y \in \overline{f(A)}$$

□

□

**Definition 10.5.**  $(X, \mathcal{T})$ . We say  $X$  is second countable iff  $X$  has a countable basis.

**Proposition 10.6.** Second  $\Rightarrow$  First

*Proof.*  $\mathcal{B} = \{B_n : n \in \mathbb{N}\}$  basis,  $\mathcal{B}_x = \{B_n : x \in B_n\}$  basis at  $x$ .

□

**Example 10.7.**  $(\mathbb{R}, P(\mathbb{R}))$  with

$$d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases}$$

Hence it is first countable. Every basis for  $\mathbb{R}$  must contain all  $\{x\}, \forall x \in \mathbb{R}$ . Thus, every basis is uncountable.

**Definition 10.8.**  $A \subseteq X, (X, \mathcal{T})$ . We say  $A$  is dense in  $X$  iff  $\overline{A} = X$ .

**Definition 10.9.**  $(X, \mathcal{T})$ . We say  $X$  is separable iff  $X$  has a countable, dense subset.

**Example 10.10.**  $\overline{\mathbb{Q}} \subseteq \mathbb{R}$

**Definition 10.11.** We say  $(X, \mathcal{T})$  is Lindelöf iff every open cover of  $X$  has a countable subcover.

**Proposition 10.12.** If  $(X, \mathcal{T})$  is second countable, then  $X$  is separable and Lindelöf

**Fact:** The lower limit topology on  $\mathbb{R}$  is separable and Lindelöf, but not second countable.

*Proof.* Let  $\mathcal{B} = \{B_n : n \in \mathbb{N}\}$  be a countable basis for  $X$ . We may assume that each  $B_n \neq \emptyset$ .

**Claim:**  $X$  is separable

Take  $x_n \in B_n$ , consider  $D = \{x_n : x_n \in \mathbb{N}\}$ . Let  $x \in X$  and let  $x \in U$  be open. Hence there exists  $n \in \mathbb{N}$  so  $x \in B_n \subseteq U$ . Thus,  $x_n \in U \cap D$  and so  $x \in \overline{D}$ . Hence  $X = \overline{D}$ .  $\square$

**Claim:**  $X$  is Lindelöf.

Suppose  $X = \bigcup U_\alpha$  is an open cover. Let  $I = \{n \in \mathbb{N} : \exists \alpha, B_n \subseteq U_\alpha\}$

**Subclaim:** Say  $B_n \subseteq U_{\alpha_n}$  for all  $n \in I$ . Then  $X = \bigcup_{n \in I} U_{\alpha_n}$ .

Take  $x \in X$ , there exists  $\alpha$  such that  $x \in U_\alpha$ . Hence there exists  $n \in \mathbb{N}$ ,  $x \in B_n \subseteq U_\alpha$ . Thus  $x \in U_{\alpha_n}$   $\square$

## 10.1 Separation

**Convention**  $(X, \mathcal{T})$ . Assume  $\forall x \in X, \{x\}$  is closed.

**Definition 10.13.** 1. We say  $(X, \mathcal{T})$  is regular iff  $\forall x \in X, \forall C \subseteq X$  closed with  $x \notin C$ , then there exists disjoint open  $U, V \in \mathcal{T}$  such that  $x \in U, C \subseteq V$

2. We say  $(X, \mathcal{T})$  is normal iff  $\forall$  disjoint, closed  $A, B \subseteq X$ , there exists disjoint  $U, V \in \mathcal{T}$  such that  $A \subseteq U$  and  $B \subseteq V$

**Remark 10.14.** Using the convention, Normal  $\implies$  Regular  $\implies$  Hausdorff

**Recall** If  $(X, \mathcal{T})$  is compact, Hausdorff, then  $X$  is regular.

**Lemma 10.15.**  $(X, \mathcal{T})$ :

1. Then,  $X$  is regular iff  $\forall x \in X, \forall x \in U \in \mathcal{T}$ , there exists  $x \in V \in \mathcal{T}$  such that  $\overline{V} \subseteq U$ .

2. Then,  $X$  is normal iff  $\forall A \subseteq X$  closed,  $\forall A \subseteq U \in \mathcal{T}$ , there exists  $A \subseteq V \in \mathcal{T}, \overline{V} \subseteq U$

*Proof.*  $(\implies)$ : Suppose that  $X$  is regular. Let  $x \in X$  and  $U$  be a nbhd of  $x$ . Consider  $C = X \setminus U$ . By regularity, there exists disjoint  $V, W \in \mathcal{T}$  such that  $x \in V, C \subseteq W$ . Suppose, for contradiction,  $z \in \overline{V}$  but  $z \notin U$ . Then  $z \in C \subseteq W$  and so  $V \cap W \neq \emptyset$ , contradiction.

$(\impliedby)$ : Let  $x \in X$  and let  $C \subseteq X$  is closed with  $x \notin C$ . Then  $U = X \setminus C$  is open with  $x \in U$ . By assumption, there exists  $x \in V \in \mathcal{T}$  such that  $\overline{V} \subseteq U$ . Let  $W = X \setminus \overline{V} \in \mathcal{T}$ . Note that  $W \cap V = \emptyset$ . Finally,  $C = X \setminus U \subseteq X \setminus \overline{V} = W$ .

The proof of (2) is similar.  $\square$

**HW:** Every metric space is normal.

**Proposition 10.16.** If  $X$  is compact and Hausdorff, then  $X$  is normal.

*Proof.* Let  $A, B$  be disjoint closed sets. Since  $X$  is regular,  $\forall x \in A, \exists x \in U_x \in \mathcal{T}$  and  $B \subseteq V_x \in \mathcal{T}$  with  $U_x \cap V_x = \emptyset$ . Then  $A \subseteq \bigcup U_x$ . Since  $A$  is compact,  $A \subseteq U_{x_1} \cup \dots \cup U_{x_n}$  for some  $x_i$ 's. Finally,

$$B \subseteq V_{x_1} \cap \dots \cap V_{x_n}$$

and indeed,  $A, B$  are disjoint. □

**Theorem 10.17** (Urysohn's Lemma).  $(X, \mathcal{T})$  normal,  $A, B$  disjoint, closed. There exists a cts  $f : X \rightarrow [0, 1]$  such that

$$f|_A = 0 \text{ and } f|_B = 1$$

*Proof.* Let  $P = \{p_1 = 1, p_2 = 0, p_3, p_4, \dots\}$  be an enumeration of  $\mathbb{Q} \cap [0, 1]$

1. For  $p, q \in P$ , we will construct  $U_p, U_q \in \mathcal{T}$  such that  $p < q \implies \overline{U_p} \subseteq U_q$   
We proceed by induction. Set  $U_1 = X \setminus B$ . By normality, there exists  $U_0 \in \mathcal{T}$  such that  $A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1$ . Suppose  $P_n = \{p_1, \dots, p_n\}$  and that  $\forall p \in P_n, \exists U_p \in \mathcal{T}$  such that

$$p, q \in P_n \text{ and } p < q \implies \overline{U_p} \subseteq U_q \quad (\dagger)$$

Let  $r = p_{n+1}$ . Say  $p, q \in P_n$  are closest such that  $p < r < q$ . Notice that  $\overline{U_p} \subseteq U_q$ . By normality, there exists  $U_r \in \mathcal{T}$  such that  $\overline{U_p} \subseteq U_r$  and  $\overline{U_r} \subseteq U_q$ .

For  $s \in P_n$ ,

$$\begin{aligned} s \leq p &\implies \overline{U_s} \subseteq \overline{U_p} \subseteq U_r \\ s \geq q &\implies \overline{U_r} \subseteq U_q \subseteq \overline{U_q} \subseteq U_s \end{aligned}$$

This proves  $(\dagger)$  for  $P_{n+1}$ . By induction, such  $U_p \in \mathcal{T}$  exists for all  $p \in P$ .

2. For  $p \in \mathbb{Q}$ , set  $U_p = \emptyset$  for  $p < 0$ , and let  $U_p = X$  for  $p > 1$ . Thus

$$p < q \implies \overline{U_p} \subseteq U_q.$$

3. For  $x \in X$ , let  $Q(x) = \{p \in \mathbb{Q} : x \in U_p\}$ . Notice that  $Q(x) \subseteq [0, \infty)$  and  $(1, \infty) \subseteq Q(x)$ . Let  $f(x) = \inf Q(x)$ .

4. Show  $f$  is cts,  $f|_A = 0, f|_B = 1$ .

If  $x \in A$ , then  $x \in U_0 \implies 0 \in Q(x)$ . Therefore,  $f(x) = 0$ . If  $x \in B$ , then for  $p < 1$ ,  $\overline{U_p} \subseteq U_1 = X \setminus B$ , and hence  $x \notin U_p$ . Therefore  $f(x) = \inf Q(x) = 1$

We must prove that  $f$  is continuous. Notice that

- $x \in \overline{U_r} \implies x \in U_s$  for  $r < s \implies f(x) \leq r$
- $x \notin U_r \implies x \notin U_s$  for  $s < r \implies f(x) \geq r$

Fix  $x_0 \in X$  with  $f(x_0) \in (c, d) \subseteq \mathbb{R}$ . Choose  $p, q \in \mathbb{Q}$  such that  $c < p < f(x_0) < q < d$ .

Let  $U = U_q \setminus \overline{U_p}$  so that  $x_0 \in U$ . It remains to prove that  $f(U) \subseteq (c, d)$ .

For  $x \in U_q \subseteq \overline{U_q} \implies f(x) \leq q$ , and  $x \notin \overline{U_p} \implies x \notin U_p \implies f(x) \geq p$ . Thus,

$$f(x) \in [p, q] \subseteq (c, d)$$

Hence  $f(U) \subseteq (c, d)$  and  $f$  is cts. □

**HW:** Extend this to  $[a, b]$

**Lemma 10.18.**  $(X, \mathcal{T})$  normal,  $A \subseteq X$  closed. If  $f : A \rightarrow [-r, r]$  is cts, then there exists  $g : X \rightarrow \mathbb{R}$  such that

1.  $\forall x \in X, |g(x)| \leq \frac{r}{3}$
2.  $\forall a \in A, |g(a) - f(a)| \leq \frac{2r}{3}$

*Proof.* Let  $I_1 = [-r, -\frac{r}{3}]$ ,  $I_2 = [-\frac{r}{3}, \frac{r}{3}]$ , and  $I_3 = [\frac{r}{3}, r]$ . Consider  $\underbrace{C = f^{-1}(I_1), D = f^{-1}(I_3)}_{\text{closed+disjoint}}$ .

By Urysohn, there exists a continuous  $g : X \rightarrow [-\frac{r}{3}, \frac{r}{3}]$  such that

$$g|_C = -\frac{r}{3}, g|_D = \frac{r}{3}$$

1.  $\forall x, |g(x)| \leq \frac{r}{3}$
2.  $\forall a \in A$ :
  - (a) if  $a \in C, f(a), g(a) \in I_1$
  - (b) if  $a \in D, f(a), g(a) \in I_3$
  - (c) if  $a \notin C \cup D, f(a), g(a) \in I_2$

and so  $|g(a) - f(a)| \leq \frac{2r}{3}$ .

□

**Theorem 10.19** (Tietze Extension Theorem).  $(X, \mathcal{T})$  normal,  $A \subseteq X$  closed. Every continuous  $f : A \rightarrow \mathbb{R}$  can be extended to a continuous  $f : X \rightarrow \mathbb{R}$ .

*Proof.* **Case 1:**  $f(A) \subseteq [-1, 1]$ .

The lemma above implies there exists  $g_1 : X \rightarrow \mathbb{R}$  such that

$$|g_1(x)| \leq \frac{1}{3}, |f(a) - g_1(a)| \leq \frac{2}{3}.$$

Hence  $f - g_1 : A \rightarrow [-\frac{2}{3}, \frac{2}{3}]$ .

Again, by lemma, there exists cts  $g_2 : X \rightarrow \mathbb{R}$  such that

$$|g_2(x)| \leq \frac{1}{3}(\frac{2}{3}), |f(a) - g_1(a) - g_2(a)| \leq (\frac{2}{3})^2.$$

Hence  $f - g_1 - g_2 : A \rightarrow [-\frac{2}{3}(\frac{2}{3}), \frac{2}{3}(\frac{2}{3})]$ .

Doing this inductively, we have for all  $n \in \mathbb{N}$ , there exists cts  $g_n : X \rightarrow \mathbb{R}$  such that

$$|g_n(x)| \leq \frac{1}{3}(\frac{2}{3})^{n-1}, |f(a) - g_1(a) - \dots - g_n(a)| \leq (\frac{2}{3})^n.$$

Consider  $g : X \rightarrow \mathbb{R}$  with  $g(x) = \sum_{n=1}^{\infty} g_n(x)$ , which converges by comparison with

$$\sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{2}{3}\right)^{n-1} = \frac{\frac{1}{3}}{1 - \frac{2}{3}} = 1.$$

By the M-test,  $g$  is cts.

For all  $a \in A$ ,  $|f(a) - \sum_{i=1}^n g_i(a)| \leq \left(\frac{2}{3}\right)^n \rightarrow 0 \implies f(a) = g(a)$ . Note:  $g : X \rightarrow [-1, 1]$ .

**Case 2:**  $f(A) \subseteq (-1, 1)$ .

By Case 1, we can extend  $f$  to a cts  $g : X \rightarrow [-1, 1]$ . Consider  $D = g^{-1}(\{-1, 1\})$  is closed. Then:

$$g(A) = f(A) \subseteq (-1, 1) \implies A \cap D = \emptyset.$$

By Urysohn, there exists a cts  $\varphi : X \rightarrow [0, 1]$  such that  $\varphi|_D = 0$  and  $\varphi|_A = 1$ . Consider:

$$\begin{aligned} h : X &\rightarrow (-1, 1) \text{ with } h(x) = \varphi(x)g(x) \\ \implies \forall a \in A, h(a) &= \varphi(a)g(a) = g(a) = f(a) \end{aligned}$$

**Case 3:**  $f : A \rightarrow \mathbb{R}$ .

Let  $\psi : \mathbb{R} \rightarrow (-1, 1)$  be a homeomorphism. Then  $\psi \circ f : A \rightarrow (-1, 1)$  is cts. By Case 2, we may extend  $\psi \circ f$  to a cts  $g : X \rightarrow (-1, 1)$ . Consider  $\psi^{-1} \circ g : X \rightarrow \mathbb{R}$ . For all  $a \in A$ :

$$\psi^{-1}(g(a)) = \psi^{-1}(\psi(f(a))) = f(a).$$

□

## 11 Topological Manifold

**Definition 11.1.** For  $m \in \mathbb{N}$ , an  $m$ -manifold is a Hausdorff, second-countable space  $(X, \mathcal{T})$  such that

$$\forall x \in X, \exists x \in U \in \mathcal{T}, U \cong \text{open subset of } \mathbb{R}^m$$

**Definition 11.2.**  $(X, \mathcal{T}), (Y, \mathcal{U})$ . An **embedding** of  $X$  into  $Y$  is a continuous injective  $f : X \rightarrow Y$  such that  $f^{-1} : f(X) \rightarrow X$  is also continuous. i.e.,  $f : X \rightarrow f(X)$  is a homeomorphism.

**Theorem 11.3.** If  $X$  is a compact  $m$ -manifold, then  $X$  can be embedded in  $\mathbb{R}^N$  for some  $N$ .

**Definition 11.4.**  $(X, \mathcal{T}), f : X \rightarrow \mathbb{R}$ . The support of  $f$  is  $\text{supp}(f) = \{x \in X : f(x) \neq 0\}$

**Definition 11.5.**  $(X, \mathcal{T})$ . Let  $U_1, \dots, U_n$  be a finite open cover of  $X$ . A partition of unity dominated by  $\{U_i\}$  is the family of cts

$$\varphi_i : X \rightarrow [0, 1], \quad i = 1, 2, \dots, n$$

such that

1.  $\text{supp}(\varphi_i) \subseteq U_i$

$$2. \forall x, \sum \varphi_i(x) = 1$$

**Proposition 11.6.**  $(X, \mathcal{T})$  normal. If  $U_1, \dots, U_n$  is a finite open cover of  $X$ , then exists a partition of unity dominated by  $U_i$ .

*Proof.* First, we can find an open cover  $v_1, \dots, v_n$  of  $X$  such that  $\overline{v_i} \subseteq U_i$ . To start, let

$$A = \underbrace{X \setminus U_2 \cup \dots \cup U_n}_{\text{closed}} \subseteq U_1$$

By normality lemma, there exists  $V_1 \in \mathcal{T}$  such that  $A \subseteq V_1 \subseteq \overline{V_1} \subseteq U_1$ . Note that  $V_1, U_2, \dots, U_n$  covers  $X$ . Inductively, assume  $V_1, \dots, V_{k-1}, U_k, \dots, U_n$  covers  $X$  with  $\overline{V_i} \subseteq U_i$ . Let

$$A = X \setminus (v_1 \cup \dots \cup V_{k-1} \cup U_{k+1} \cup \dots \cup U_n) \subseteq U_k$$

Similarly, we find  $V_k \in \mathcal{T}$  such that  $A \subseteq V_k \subseteq \overline{V_k} \subseteq U_k$  and  $V_1, \dots, V_k, U_{k+1}, \dots, U_n$  covers  $X$ . By induction, such a cover  $V_1, V_2, \dots, V_n$  exists.

Once again, let  $W_1, \dots, W_n$  be a cover of  $X$  such that

$$\overline{W_i} \subseteq V_i$$

Note that  $\overline{W_i}, X \setminus V_i$  are disjoint, closed sets. By Urysohn, there exists cts

$$\psi_i : X \rightarrow [0, 1], \psi_i|_{\overline{W_i}} = 1, \psi_i|_{X \setminus V_i} = 0$$

hence  $\text{supp}(\psi_i) \subseteq \overline{W_i} \subseteq V_i$ .

Since  $W_i$  cover  $X$ , we have  $\sum \psi_i(x) > 0$ . Consider

$$\varphi_i : X \rightarrow [0, 1], \varphi_i(x) = \frac{\psi_i(x)}{\sum \psi_j(x)}.$$

Hence  $\text{supp}(\varphi_i) = \text{supp}(\psi_i) \subseteq U_i$  and  $\sum \varphi_i(x) = 1$  □

*of Theorem 11.3.* By compactness, there exists an open cover  $U_1, \dots, U_n$  of  $X$  and embeddings  $g_i : U_i \rightarrow \mathbb{R}^m$ . Since  $X$  is normal, let  $\varphi_1, \dots, \varphi_n$  be a partition of unity dominated by  $\{U_i\}$ . Let  $A_i = \text{supp}(\varphi_i) \subseteq U_i$ . Consider:

$h_i : X \rightarrow \mathbb{R}^m$  given by

$$h_i(x) = \begin{cases} \varphi_i(x)g_i(x), & x \in U_i \\ 0, & x \in X \setminus A_i \end{cases}$$

Note that if  $x \in U_i \cap X \setminus A_i$ , then  $\varphi_i(x) = 0$ . By a lemma,  $h_i|_{U_i}, h_i|_{X \setminus A_i}$  are continuous  $\implies h_i$  is cts.

Define  $F : X \rightarrow \underbrace{\mathbb{R} \times \dots \times \mathbb{R}}_{n\text{-times}} \times \underbrace{\mathbb{R}^m \times \dots \times \mathbb{R}^m}_{n\text{-times}}$ ,  $F(x) = (\varphi_1(x), \dots, \varphi_n(x), h_1(x), \dots, h_n(x))$ . For all  $i$ ,  $\pi_i \circ F$  is cts and thus  $F$  is cts. Since  $X$  is compact, we need only prove that  $F$  is injective. Suppose  $x, y \in X$

such that  $F(x) = F(y)$ . Since  $\sum \varphi_i(x) = 1, \exists i, \varphi_i(x) > 0 \implies \varphi_i(x) = \varphi_i(y) > 0 \implies x, y \in A_i \subseteq U_i$ . Thus, for that same  $i$ ,

$$\begin{aligned} h_i(x) &= h_i(y) \\ \implies \varphi_i(x)g_i(x) &= \varphi_i(y)g_i(y) \\ \implies g_i(x) &= g_i(y) \\ \implies x &= y. \end{aligned}$$

□

**Facts:**

1.  $X$  compact  $m$ -manifold, then  $X \hookrightarrow \mathbb{R}^{2m+1}$ .
2.  $X$  compact, smooth,  $m$ -manifold,
  - (a)  $X \hookrightarrow \mathbb{R}^{2m}$
  - (b)  $\exists X, X \not\hookrightarrow \mathbb{R}^{2m-1}$

## 12 The Fundamental Group

**Goal:** Classify spaces (surfaces) in terms of how paths in the spaces behave. We will introduce algebraic concepts to deal with hard geometry.

**Idea:** Try to continuously deform one path into another.

**Definition 12.1.**  $(X, \mathcal{T})$ ,  $x_0, x_1 \in X$ . A homotopy of paths in  $X$  is a family

$$f_t : [0, 1] \rightarrow X, t \in [0, 1]$$

such that

1.  $\forall t, f_t(0) = x_0$  and  $f_t(1) = x_1$ .
2.  $F : [0, 1] \times [0, 1] \rightarrow X$ ,  $F(s, t) = f_t(s)$  is continuous.

**Remark 12.2.** Say  $s_i \rightarrow s$  in  $[0, 1]$

$$\begin{aligned} &\implies \forall t, F(s_i, t) \rightarrow F(s, t) \\ &\implies \forall t, f_t(s_i) \rightarrow f_t(s) \\ &\implies \forall t, f_t \text{ is continuous} \\ &\implies \forall t, f_t \text{ is a path} \end{aligned}$$

**Definition 12.3.**  $(X, \mathcal{T})$ . Let  $f, g : [0, 1] \rightarrow X$  be paths with  $f(0) = g(0) = x_0$  and  $f(1) = g(1) = x_1$ . We say  $f$  and  $g$  are homotopic,  $f \simeq g$  iff there exists homotopy (of paths)  $f_t$  such that

$$f_0 = f \text{ and } f_1 = g$$

**Example 12.4.**  $X \subseteq \mathbb{R}^n$  convex.  $f_0, f_1$  paths in  $X$  (some endpoints).  $f_t(s) = (1-t)f_0(s) + tf_1(s)$

$$\implies f_0 \simeq f_1$$

**Proposition 12.5.**  $(X, \mathcal{T})$ ,  $x_0, x_1 \in X$ .  $P = \{f : [0, 1] \rightarrow X \mid f \text{ continuous}, f(0) = x_0, f(1) = x_1\}$ . Then,  $\simeq$  is an equivalence relation on  $P$ .

*Proof.* First, let  $f \in P$ . Then  $f_t = f$  is a homotopy from  $f$  to  $f$ , hence  $f \simeq f$ .

Second, suppose  $f \simeq g$  via  $f_t$ . Then  $g \simeq f$  via  $f_{1-t}$ .

Last, suppose  $f \simeq g$  via  $f_t$  and  $g \simeq h$  via  $g_t$ , consider

$$h_t(s) = \begin{cases} f_{2t}(s), & 0 \leq t \leq \frac{1}{2} \\ g_{2t-1}(s), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

Note that  $h_{\frac{1}{2}}(s) = f_1(s) = g(s)$ , and  $h_{\frac{1}{2}} = g_0(s) = g(s)$ . Then  $h_t$  is a homotopy from  $h_0 = f$  to  $h_1 = h$ . □

**Notation:** We denote  $[f]$  to be the homotopy class of  $f$ .

With  $(X, \mathcal{T})$ ,  $f, g : [0, 1] \rightarrow X$  cts,  $f(1) = g(0)$ , then we define the product path to be

$$f \cdot g : [0, 1] \rightarrow X$$

$$(f \cdot g)(s) = \begin{cases} f(2s), & 0 \leq s \leq \frac{1}{2} \\ g(2s - 1), & \frac{1}{2} \leq s \leq 1 \end{cases}$$

In A5: IF  $f_0 \simeq f_1$  via  $f_t$  and  $g_0 \simeq g_1$  via  $g_t$ , then  $f_0 \cdot g_0 \simeq f_1 \cdot g_1$  via  $f_t \cdot g_t$ .

**Definition 12.6.**  $(X, \mathcal{T})$ . A loop in  $X$  is a path  $f : [0, 1] \rightarrow X$  such that  $f(0) = f(1)$ .

**Definition 12.7.**  $(X, \mathcal{T})$ ,  $x_0 \in X$ . We call

$$\pi_1(X, x_0) = \{[f] : f : [0, 1] \rightarrow X \text{ loop}, f(0) = f(1) = x_0\}$$

the fundamental group of  $X$  at the basepoint  $x_0$ .

**Proposition 12.8.**  $\pi_1(X, x_0)$  is a group via  $[f][g] = [f \cdot g]$ .

**Remark 12.9.** By A5, this operation is well-defined.

**Main Tool:** Reparametrization

$f : [0, 1] \rightarrow X$  path,  $\varphi : [0, 1] \rightarrow [0, 1]$  continuous,  $\varphi(0) = 0, \varphi(1) = 1$ . Then  $f \circ \varphi$  is a path with endpoints  $f(0), f(1)$  such that  $f \circ \varphi \simeq f$ .

Why?  $f \circ \varphi_t, \varphi_t(s) = (1 - t)\varphi(s) + ts$ , which gives the homotopy.

*Proof.* First, we want to show the associativity. Let  $[f], [g], [h] \in \pi_1(X, x_0)$ , then

$$(f \cdot g) \cdot h(s) = \begin{cases} f \cdot g(2s) & 0 \leq s \leq \frac{1}{2} \\ h(2s - 1) & \frac{1}{2} \leq s \leq 1 \end{cases} = \begin{cases} f(4s) & 0 \leq s \leq \frac{1}{4} \\ g(4s - 1) & \frac{1}{4} \leq s \leq \frac{1}{2} \\ h(2s - 1) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

□

and

$$f \cdot (g \cdot h)(s) = \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ g \cdot h(2s - 1) & \frac{1}{2} \leq s \leq 1 \end{cases} = \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ g(4s - 2) & \frac{1}{2} \leq s \leq \frac{3}{4} \\ h(4s - 3) & \frac{3}{4} \leq s \leq 1 \end{cases}$$

consider  $\varphi : [0, 1] \rightarrow [0, 1]$ , then

$$\varphi(s) = \begin{cases} \frac{1}{2}s & 0 \leq s \leq \frac{1}{2} \\ s - \frac{1}{4} & \frac{1}{2} \leq s \leq \frac{3}{4} \\ 2s - 1 & \frac{3}{4} \leq s \leq 1 \end{cases}$$

Thus,  $((f \cdot g) \cdot h) \circ \varphi = f \cdot (g \cdot h) \implies (f \cdot g) \cdot h \simeq f \cdot (g \cdot h)$

Second, we want to verify the identity. Let  $c$  be the constant loop  $c(s) = x_0$  and let  $[f] \in \pi_1(X, x_0)$ , then

$$f \cdot c(s) = \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ x_0 & \frac{1}{2} \leq s \leq 1 \end{cases}$$

Consider  $\varphi : [0, 1] \rightarrow [0, 1]$ , then

$$\varphi(s) = \begin{cases} 2s & 0 \leq s \leq \frac{1}{2} \\ 1 & \frac{1}{2} \leq s \leq 1 \end{cases}$$

Indeed,  $f \circ \varphi = f \cdot c \implies f \simeq f \cdot c$ .

Next, consider the other way:

$$c \cdot f(s) = \begin{cases} x_0 & 0 \leq s \leq \frac{1}{2} \\ f(2s-1) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

Consider  $\varphi : [0, 1] \rightarrow [0, 1]$ ,

$$\varphi(s) = \begin{cases} 0 & 0 \leq s \leq \frac{1}{2} \\ 2s-1 & \frac{1}{2} \leq s \leq 1 \end{cases}$$

Hence  $f \circ \varphi = c \cdot f \implies c \cdot f$ . Indeed, the identity is  $[c]$ .

Lastly, we want to verify the inverse. Let  $[f] \in \pi_1[X, x_0]$  and consider  $\bar{f}(s) = f(1-s)$ , then

$$f \cdot \bar{f}(s) = \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ \bar{f}(2s-1) & \frac{1}{2} \leq s \leq 1 \end{cases} = \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ f(2-2s) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

Consider

$$f_t(s) = \begin{cases} f(s) & s \in [0, 1-t] \\ f(1-s) & s \in [1-t, 1] \end{cases}$$

$$g_t(s) = f_t(1-s)$$

Then,  $h(t) = f_t \cdot g_t$  is a homotopy of paths from

$$\begin{aligned} h_0(s) &= f_0 \cdot g_0(s) \\ &= \begin{cases} f_0(2s) & 0 \leq s \leq \frac{1}{2} \\ g_0(2s-1) & \frac{1}{2} \leq s \leq 1 \end{cases} \\ &= \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ \bar{f}(2s-1) & \frac{1}{2} \leq s \leq 1 \end{cases} \\ &= f \cdot \bar{f} \end{aligned}$$

to

$$\begin{aligned} h_1(s) &= f_1 \cdot g_1(s) \\ &= \begin{cases} f_1(2s) & 0 \leq s \leq \frac{1}{2} \\ g_1(2s-1) & \frac{1}{2} \leq s \leq 1 \end{cases} \end{aligned}$$

TO BE FIXED HERE

**Example 12.10.**  $X \subseteq \mathbb{R}^n$  convex,  $x_0 \in X$ ,  $\pi_1(X, x_0) = \{[c]\} = \{1\}$

**Question:** How does the choice of  $x_0$  affect  $\pi_1(X, x_0)$ ?

**HW:** If  $h$  is a path from  $x_0$  to  $x_1$ , then

$$h\bar{h} \simeq 1 \simeq \bar{h}h$$

**Proposition 12.11.**  $(X, \mathcal{T})$ . Let  $h : [0, 1] \rightarrow X$  be a path from  $x_0$  to  $x_1$ . Then

$$\varphi_h : \pi_1(X, x_1) \rightarrow \pi_1(X, x_0), [f] \mapsto [h \cdot f \cdot \bar{h}]$$

is an isomorphism

*Proof.* For  $[f], [g] \in \pi_1(X, x_1)$ ,

$$\begin{aligned} \varphi_h([f \cdot g]) &= [h \cdot f \cdot g \cdot \bar{h}] \\ &= [h \cdot f \cdot \bar{h} \cdot h \cdot g \cdot \bar{h}] \\ &= [h \cdot f \cdot \bar{h}] \cdot [h \cdot g \cdot \bar{h}] \\ &= \varphi_h([f]) \cdot \varphi_h([g]) \end{aligned}$$

**Claim:**  $(\varphi_h)^{-1} = \varphi_{\bar{h}}$

*claim.*  $\varphi_h \circ \varphi_{\bar{h}}([f]) = \varphi_n([\bar{h} \cdot f \cdot h]) = [h \cdot \bar{h} \cdot f \cdot h \cdot \bar{h}] = [f]$ . Similarly,  $\varphi_{\bar{h}} \circ \varphi_h = \text{id}$ . Hence we conclude that  $\varphi_h$  is an isomorphism. □

□

**HW:** If  $f$  is a path from  $x_0$  to  $x_1$  and  $c = x_0$  and  $d = x_1$  are constant loops, then

$$c \cdot f \simeq f \text{ and } f \cdot d \simeq f$$

.

**Corollary 12.12.** If  $X$  is path-connected, then  $\forall x_0, x_1 \in X$ :

$$\underbrace{\pi_1(X, x_0) \cong \pi_1(X, x_1)}_{\pi_1(X)}$$

**Definition 12.13.**  $(X, \mathcal{T})$ . We say  $X$  is simply connected iff  $X$  is path connected and  $\varphi_1(X) = \{1\}$ .

**Example 12.14.**  $D^2 = \{(x, y) : x^2 + y^2 \leq 1\}$  is simply connected.

**Example 12.15.** Convex implies simply connected.

**Proposition 12.16.**  $(X, \mathcal{T})$ . Then  $X$  is simply connected iff there is a unique homotopy class of paths connecting any two  $x_0, x_1 \in X$ .

*Proof.*  $(\Leftarrow)$ : Trivial.

$(\Rightarrow)$ : Now suppose  $X$  is simply connected. Fix  $x_0, x_1 \in X$  and paths  $f, g$  from  $x_0$  to  $x_1$ . Then:

$$[f \cdot \bar{g}] = [1] \text{ and } [\bar{g} \cdot g] = [1]$$

hence  $f \simeq f \cdot (\bar{g} \cdot g) \simeq (f \cdot \bar{g}) \cdot g \simeq g$  □

**Sad Fact:** Computing  $\pi_1(x)$  is hard.

**Conjecture:**

$S^1 = \{(x, y) : x^2 + y^2 = 1\}$ , let  $w : [0, 1] \rightarrow S^1$  where  $w(s) = (\cos(2\pi s), \sin(2\pi s))$ . Then we might see the homotopy class in it are  $\{\bar{w}, w^2, w^{-3}, 1, \dots\}$ .

**Claim:**  $\pi_1(S^1) = \langle [w] \rangle \cong \mathbb{Z}$ .

**Big Idea:** We compare paths in  $S^1$  with paths in  $\mathbb{R}$  using

$$p : \mathbb{R} \rightarrow S^1, s \mapsto (\cos(2\pi s), \sin(2\pi s))$$

**Theorem 12.17.**  $\varphi : \mathbb{Z} \rightarrow \pi_1(S^1), \varphi(n) = [w_n]$ , where  $w_n(s) = (\cos(2\pi ns), \sin(2\pi ns))$  is an isomorphism.

[pic here] We will “lift” paths in  $S^1$  to paths in  $\mathbb{R}$ . Consider:

$$p : \mathbb{R} \rightarrow S^1, s \mapsto (\cos(2\pi s), \sin(2\pi s))$$

$$w_n = p \circ \tilde{w}_n$$

$$\tilde{w}_n : [0, 1] \rightarrow \mathbb{R}, s \mapsto ns$$

We say  $\tilde{w}_n$  is a lift of  $w_n$ .

*Proof.* **Claim:**  $\varphi$  is a homomorphism.

First, if  $\tilde{f} : [0, 1] \rightarrow \mathbb{R}$  is a path from 0 to  $n$ , then  $\tilde{f} \simeq \tilde{w}_n$  ( $\mathbb{R}$  convex). Thus

$$\varphi(n) = [w_n] = [p \circ \tilde{w}_n] = [p \circ \tilde{f}].$$

Consider  $\tau_m : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x + m$  ( $m \in \mathbb{Z}$ ). Then  $\tilde{w}_m \cdot (\tau_m \circ \tilde{w}_n)$  is a path in  $\mathbb{R}$  from 0 to  $m + n$ . So

$$\varphi(m + n) = [p \circ (\tilde{w}_m \cdot (\tau_m \circ \tilde{w}_n))]$$

Well...

$$\begin{aligned}
& p \circ (w_m \cdot (\tau_m \circ w_m))(s) \\
&= \begin{cases} p(w_m(2s)) & 0 \leq s \leq \frac{1}{2} \\ p(\tau_m((w_m)(2s-1))) & \frac{1}{2} \leq s \leq 1 \end{cases} \\
&= \begin{cases} p(w_m(2s)) & 0 \leq s \leq \frac{1}{2} \\ p((w_m)(2s-1) + m) & \frac{1}{2} \leq s \leq 1 \end{cases} \\
&= \begin{cases} p(2ms) & \\ p(2ns - n + m) & \end{cases} \\
&= \begin{cases} p(2ms) & \\ p(2ns - n) & \end{cases}
\end{aligned}$$

Also,

$$\begin{aligned}
w_m \cdot w_n(s) &= \begin{cases} w_m(2s) \\ w_n(2s-2) \end{cases} \\
&= \begin{cases} p(w_m(2s)) \\ p(w_n(2s-1)) \end{cases} \\
&= \begin{cases} p(2ms) \\ p(2ns - n) \end{cases}
\end{aligned}$$

Hence we conclude  $\varphi(m+n) = [w_m \cdot w_n] = [w_m][w_n] = \varphi(m)\varphi(n)$ . □

**Proposition 12.18.**  $(X, \mathcal{T}), (Y, \mathcal{U})$ .  $\pi_1(X \times Y, (x, y)) \cong \pi_1(X, x) \times \pi_1(Y, y)$ .

*Proof.* Consider  $\varphi : \varphi_1(X \times Y, (x, y)) \rightarrow \pi_1(X, x) \times \pi_1(Y, y)$  given by  $\varphi([f]) = [g] \times [h]$ , where

$$f : [0, 1] \rightarrow X \times Y, f(s) = (g(s), h(s)), g = \pi_1 \circ f, h = \pi_2 \circ f$$

First, let us verify this is a homomorphism:

Let  $[f_1], [f_2] \in \pi_1(X \times Y, (x, y))$ . Say  $f_1 = (g_1, h_1), f_2 = (g_2, h_2)$ , hence  $f_1 \cdot f_2 = (g_1 \cdot g_2, h_1 \cdot h_2)$ .

Thus  $\varphi([f_1][f_2]) = ([g_1 \cdot g_2], [h_1 \cdot h_2]) = ([g_1, h_1])([g_2, h_2]) = \varphi([f_1])$ .

Also, the surjectivity is easy. We remain to verify the injectivity:

$$\begin{aligned}
& \varphi([f_1]) = \varphi([f_2]) \\
& \implies ([g_1], [h_1]) = ([g_2], [h_2]) \\
& \implies g_1 \simeq g_2 \text{ via } g_t, h_1 \simeq h_2 \text{ via } h_t
\end{aligned}$$

Thus,  $f_1 \simeq f_2$  via  $f_t(s) = (g_t(s), h_t(s)) \implies [f_1] = [f_2]$ . □

**Theorem 12.19** (Fundamental Theorem of Algebra). Every non-constant  $p(z) \in \mathbb{C}[z]$  has a root

*Proof.* Suppose  $p(z) = z^n + a_{n-1}z^{n-1} + \dots + a_1z + a_0$  has no root in  $\mathbb{C}$ . For  $r \geq 0$ ,

$$f_r(s) = \frac{p(re^{2\pi is})/p(r)}{|p(re^{2\pi is})/p(r)|}$$

is a family of loops at 1 in  $S^1 \subseteq \mathbb{C}$ . Note that  $f_0(s) = 1$ , hence  $[f_r] = [1]$ .

Say  $\pi_1(S^1) = \langle w \rangle \cong \mathbb{Z}$ . Take  $r > 1$  with

$$r > |a_{n-1}| + \dots + |a_1| + |a_0|$$

For  $z \in \mathbb{C}$  with  $|z| = r$ , then

$$\begin{aligned} |z|^n &= r^n = r \cdot r^{n-1} \\ &> (|a_{n-1}| + \dots + |a_1| + |a_0|)|z|^{n-1} \\ &\geq |a_{n-1}z^{n-1} + \dots + a_1z + a_0| \end{aligned}$$

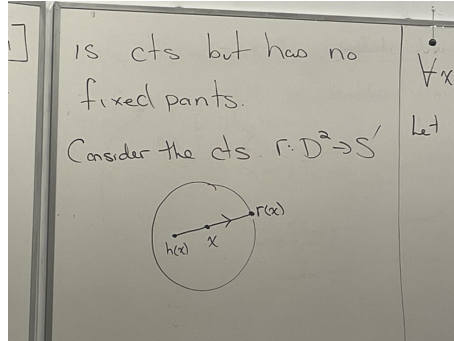
For  $0 \leq t \leq 1$ , consider  $p_t(z) = z^n + t(a_{n-1}z^{n-1} + \dots + a_1z + a_0)$ , which has no roots in the circle  $|z| = r$ . Then

$$\frac{p_t(re^{2\pi is})/p_t(r)}{|p_t(re^{2\pi is})/p_t(r)|}$$

provides a homotopy between  $f_r$  and  $w^n$ . Hence  $[w]^n = [f_r] = [1]$ , which can only happen when  $n = 0$ . Thus, the only polynomial in  $\mathbb{C}[x]$  with no root is constant, as desired.  $\square$

**Theorem 12.20** (Brouwer Fixed Point Theorem). Every continuous  $h : D^2 \rightarrow D^2$  has a fixed point.

*Proof.* Suppose  $h : D^2 \rightarrow D^2$  is continuous but has no fixed point. Consider the continuous  $r : D^2 \rightarrow S^1$  given by



$\forall x \in S^1, r(x) = x$ . Let  $[f_0] \in \pi_1(S^1)$ , hence there exists a homotopy  $f_t$  in  $D^2$  from  $f_0$  to  $f_1 = x_0$  (basepoint).

By A5,  $r \circ f_t$  is a homotopy in  $S^1$  from  $r \circ f_0 = f_0$  to  $r \circ f_1 = x_0$ . Thus,  $[f_0] = [1]$ , contradiction.  $\square$

**Goal:** Compute  $\pi_1(x)$  when  $X = A \cup B$  and  $\pi_1(A), \pi_1(B)$  are easier to compute.

**Main Tool:** Free products of groups. For each  $\alpha$ , let  $G_\alpha$  be a group. The free products of the  $G_\alpha$ 's is

$$*_\alpha G_\alpha = \{g_1 g_2 \cdots g_m : m \geq 0, g_i \in G_{\alpha_i}, \alpha_i \neq \alpha_{i+1}\}$$

The group operation is

$$\begin{aligned} & (g_1 g_2 \cdots g_m)(h_1 h_2 \cdots h_n) \\ &= g_1 g_2 \cdots g_m h_1 h_2 \cdots h_n \\ & \text{and reduce} \end{aligned}$$

**Example 12.21.**  $a_i, b_i \in G, i = 1, 2, 3$ . Then

$$\begin{aligned} & (a_1 a_2 a_3)(a_3^{-1} b_2 b_1 b_3) \\ &= a_1 a_2 a_3 a_3^{-1} b_2 b_1 b_3 \\ &= a_1 (a_2 b_2) b_1 b_3 \end{aligned}$$

**Remark 12.22.**

**Example 12.23.**  $G = \langle x \rangle \cong C_2, H = \langle y \rangle \cong C_2, G * H = \langle x, y : x^2 = y^2 = 1 \rangle$ .

**Example 12.24.**  $\mathbb{Z} \cong \langle x \rangle = \{x^n : n \in \mathbb{Z}\}$  is the free group on one generator.

**Example 12.25.**  $G = \langle x \rangle \cong \mathbb{Z}, H = \langle y \rangle \cong \mathbb{Z}$ , then  $G * H = \langle x, y \rangle$  is the free group on two generators.

**Example 12.26.**  $G = \langle x \rangle \cong \mathbb{Z}, H = \langle y \rangle \cong C_4$ , then  $G * H = \langle x, y : y^4 = 1 \rangle$

**Example 12.27.**  $G = \langle x \rangle \cong C_2, H = D_4 = \langle r, s : r^4 = s^2 = 1, rsr = s \rangle$ , then

$$G * H = \langle x, r, s : x^2 = r^4 = s^2 = 1, rsr = s \rangle.$$

**Remark 12.28.** Let  $\varphi_\alpha : G_\alpha \rightarrow H$  be a homomorphism for every  $\alpha$ . These homomorphisms extend uniquely to a homomorphism  $\varphi : *_\alpha G_\alpha \rightarrow H$  given by

$$\varphi(g_1 \cdots g_n) = \varphi_{\alpha_1}(g_1) \cdots \varphi_{\alpha_n}(g_n), g_i \in G_{\alpha_i}$$

**Theorem 12.29** (Van Kampen's Theorem). Suppose  $X = A \cup B$ , where  $A, B \in \mathcal{T}$  are path connected and  $X_0 \in A \cap B \neq \emptyset$  is also path connected. Then

1. The natural hom's:

$$\begin{aligned} i_1 : \pi_1(A) &= \pi_1(X), \\ i_2 : \pi_1(B) &= \pi_1(X) \end{aligned}$$

induce a surjective hom

$$\Phi : \pi_1(A) * \pi_1(B) \rightarrow \pi_1(X)$$

2. If  $N = \ker(\Phi)$ , then

$$\pi_1(A) * \pi_1(B) / N \cong \pi_1(X)$$

Moreover,  $N$  is the normal subgroup generated by

$$j_1(w)j_2(w)^{-1}, w \in \pi_1(A \cap B)$$

[think  $\overline{j_1(w)} = \overline{j_2(w)}$ ], where

$$\begin{aligned} j_1 : \pi_1(A \cap B) &= \pi_1(A), \\ j_2 : \pi_1(A \cap B) &= \pi_1(B) \end{aligned}$$

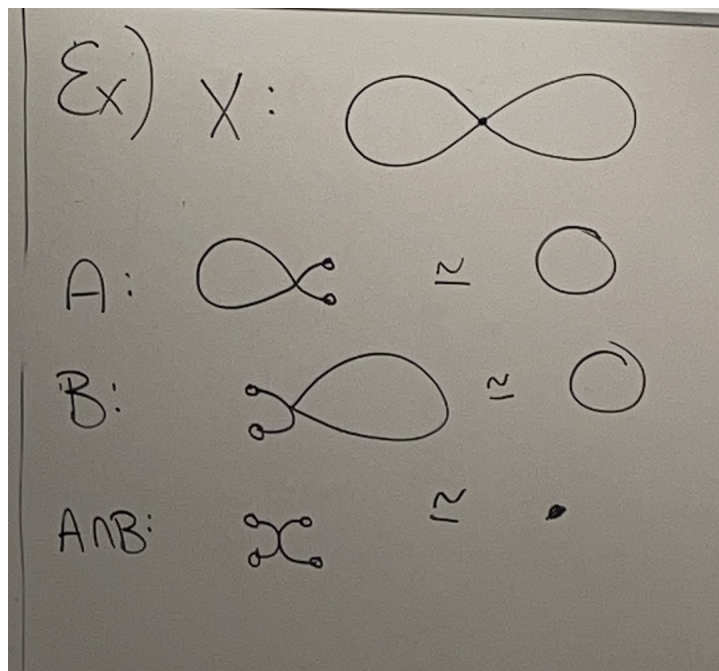
are the natural hom's

**Corollary 12.30.**  $X = A \cup B$ ,  $A, B \in \mathcal{T}$  are path connected,  $A \cap B$  is non-empty, simply connected. Then  $\pi_1(X) \cong \pi_1(A) * \pi_1(B)$ .

**Example 12.31.**  $X = S^n, n \geq 2$ . Fix  $x \in X$ .  $A = X \setminus \{x\}, B = X \setminus \{-x\}$ . Then  $A, B$  are path connected and  $A \cap B \neq \emptyset$  is path connected.

By A6,  $A, B \cong \mathbb{R}^n$ . Then  $\pi_1(A), \pi_1(B)$  are trivial, and hence  $\pi_1(X) = \{1\} \implies X$  is simply connected.

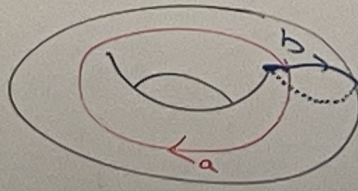
**Example 12.32.** Consider



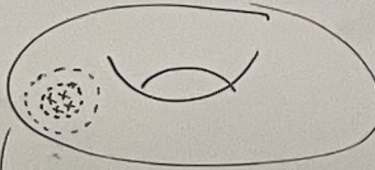
So  $\pi_1(X) \cong \pi_1(A) * \pi_1(B) \cong \mathbb{Z} * \mathbb{Z}$

**Example 12.33.** Consider

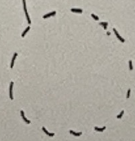
$\varepsilon_x) \times$



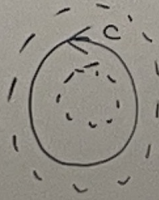
A



B



$A \cap B$



## 13 Covering Spaces

**Big Idea:** Algebraic info of  $\pi_1(x)$  should translate to geometric info of  $X$ . We obtain a Galois-like correspondence.

**Definition 13.1.**  $(X, \mathcal{T})$ . Then a covering space of  $X$  is a space  $\tilde{X}$  together with a continuous  $p: \tilde{X} \rightarrow X$  such that:

1.  $\exists$  open cover  $\{U_\alpha\}$  of  $X$  such that  $\forall \alpha, p^{-1}(U_\alpha)$  is a disjoint union of open  $V_{\alpha_i} \subseteq \tilde{X}$ .
2.  $\forall \alpha_i, p: V_{\alpha_i} \rightarrow U_\alpha$  is a homeomorphism.

**Remark 13.2.** This just means that  $\forall x \in X, \exists x \in U_\alpha$  such that  $p^{-1}(U_\alpha)$  is a disjoint union of open sets.

**Remark 13.3.** It could be that  $p^{-1}(U_\alpha) = \emptyset$ , and so  $p$  might not be surjective.

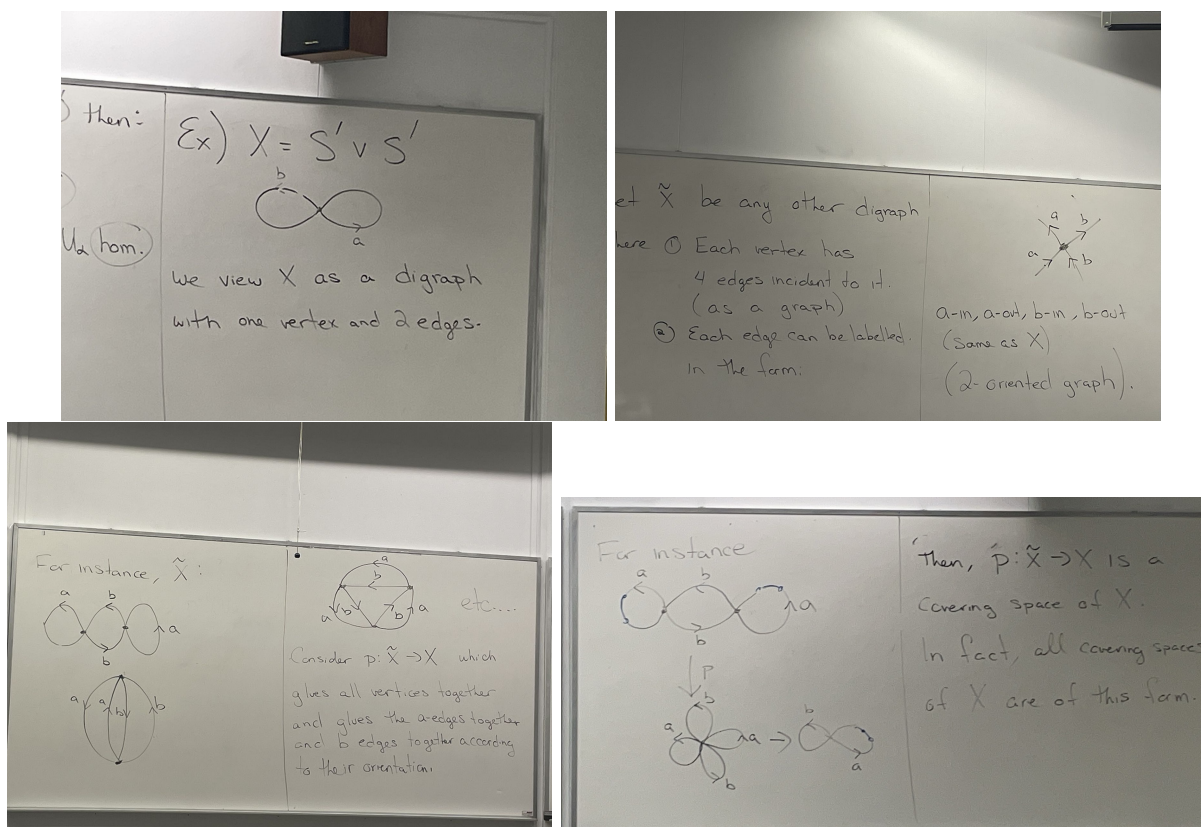
If  $\forall \alpha, p^{-1}(U_\alpha) \neq \emptyset$ , then:

$$\forall x \in X, \exists U_\alpha \text{ such that } x \in U_\alpha$$

$$p^{-1}(U_\alpha) = \bigcup V_{\alpha_i}, p: V_{\alpha_i} \rightarrow U_\alpha \text{ Hom.}$$

So there exists  $v \in V_{\alpha_i}, p(v) = x$ . Thus  $p$  is surjective.

**Example 13.4.**  $X = S^1 \vee S^1$



**Definition 13.5** (lifts).  $p : \tilde{X} \rightarrow X$ . A lift of a cts  $f : Y \rightarrow X$  is a cts  $\tilde{f} : Y \rightarrow \tilde{X}$  such that  $p \circ \tilde{f} = f$ .

**Example 13.6.**  $p : \mathbb{R} \rightarrow S^1$ ,  $p(t) = (\cos(2\pi t), \sin(2\pi t))$ .  $w_n : [0, 1] \rightarrow S^1$ ,  $w_n(t) = (\cos(2\pi t), \sin(2\pi t))$ . Then  $\tilde{w}_n : [0, 1] \rightarrow \mathbb{R}$ ,  $\tilde{w}_n(t) = nt$  is a lift of  $w_n$ .

**Proposition 13.7** (Big Fact).  $p : \tilde{X} \rightarrow X$ . Let  $F : Y \times I = [0, 1] \rightarrow X$  be cts such that  $\tilde{F} : Y \times \{0\} \rightarrow \tilde{X}$  lifts  $F : Y \times \{0\} \rightarrow X$ . Then there exists a unique lift  $\tilde{F}$  of  $F$ .

*Proof.* Page 31 of Hatcher. □

**Proposition 13.8.**  $p : \tilde{X} \rightarrow X$ . If  $f_t : Y \rightarrow X$  is a homotopy and  $f_0$  lifts to  $\tilde{f}_0$ , then exists an unique homotopy  $\tilde{f}_t : Y \rightarrow \tilde{X}$  such that  $\tilde{f}_t$  lifts each  $f_t$ .

*Proof.*  $F(s, t) = f_t(s)$ . By Big Fact, done. □

**Corollary 13.9.**  $p : \tilde{X} \rightarrow X$ . Let  $f : I \rightarrow X$  be a path starting at  $x_0$ . Suppose  $p(\tilde{x}_0) = x_0$ . Then there is a unique path  $\tilde{f} : I \rightarrow \tilde{X}$  which lifts  $f$  and starts at  $\tilde{x}_0$ .

**Corollary 13.10.**  $p : \tilde{X} \rightarrow X$ . A lift of a constant path is constant.

**Remark 13.11.**  $p : \tilde{X} \rightarrow X$ . Suppose  $f_t : I \rightarrow X$  is a homotopy of paths and assume  $\tilde{f}_0$  lifts  $f_0$ . We know there exists a unique lift homotopy  $\tilde{f}_t : I \rightarrow \tilde{X}$ .

**Question:** But is  $\tilde{f}_t$  a homotopy of paths?

Consider  $g : I \rightarrow X$ ,  $g(t) = f_t(0)$ . Let  $\tilde{g}$  be a lift of  $g$ , then  $\tilde{g}$  is constant. In particular,  $\tilde{f}_t(0)$  is constant. Same for  $\tilde{f}_t(1)$ . So the answer is YES!

**Proposition 13.12.**  $p : \tilde{X} \rightarrow X$ ,  $p(\tilde{x}_0) = x_0$ . Then,  $p_* : \pi_1(\tilde{x}, \tilde{x}_0) \rightarrow \pi_1(x, x_0)$  is injective. Moreover,

$$p_*(\pi_1(\tilde{x}, \tilde{x}_0)) = \{[f] : \tilde{f} \text{ is a loop starting at } \tilde{x}_0\}$$

*Proof.* Claim:  $p_*$  is injective.

Suppose  $[f] \in \ker(p_*)$  so that if  $f := p \circ \tilde{f}$ , then  $[f] = [1]$ . Let  $f_t$  be a homotopy in  $X$  from  $f_0 = f$  to  $f_1 = x_0$ . Then there exists a unique  $\tilde{f}_t$  homotopy which lifts  $f_t$ . Moreover,  $\tilde{f}_t$  goes from  $\tilde{f}$  to  $\tilde{x}_0$ , which is constant

$$\implies [\tilde{f}] = [1]$$

Claim 2:  $p_*(\pi_1(\tilde{x}, \tilde{x}_0)) = \{[f] : \tilde{f} \text{ is a loop starting at } \tilde{x}_0\}$ .

Take  $[f] \in \pi_1(x, x_0)$  such that  $f$  lifts to  $\tilde{f}$ , where  $\tilde{f}$  is a loop at  $\tilde{x}_0$ .

$$\implies [f] = [p \circ \tilde{f}] = p_*([\tilde{f}])$$

Finally, assume  $[g] = p_*([\tilde{f}])$ , then  $[g] = [p \circ \tilde{f}]$ , hence  $[g] = [f]$ ,

$$\implies [f] \in \pi_1(x, x_0) \text{ and its lift } \tilde{f} \text{ is a loop at } \tilde{x}_0.$$

□

**Remark 13.13.**  $p : \tilde{X} \rightarrow X$ . Let  $x \in X$  and take  $x \in U$  such that  $p^{-1}(U) = \bigcup_{i \in I} V_i$ . We know  $p : V_i \rightarrow U$  is a homeomorphism. Then  $\forall y \in U$ , there exists a unique  $x_i \in V_i$ ,  $p(x_i) = y_i$ . Hence  $|p^{-1}(\{y\})| = |I|$ . So  $\varphi : X \rightarrow \mathbb{N} \cup \{0, \infty\}$  with  $\varphi(x) = |p^{-1}(\{x\})|$  is locally constant.

Recall that  $p : \tilde{X} \rightarrow X$ ,

$$p(\tilde{x}_0) = x_0 \begin{cases} G = \pi_1(X, x_0) \\ H = \pi_1(\tilde{X}, \tilde{x}_0) \end{cases}$$

Hence  $H = \{[f] \in \pi_1(X, x_0) : \text{the unique lift } \tilde{f} \text{ starting at } \tilde{x}_0 \text{ is a loop}\}$ ,  $H \cong \pi_1(\tilde{X}, \tilde{x}_0)$

**Proposition 13.14.**  $p : \tilde{X} \rightarrow X$ ,  $X, \tilde{X}$  path-connected.  $G = \pi_1(X, x_0)$ ,  $H = p_*\pi_1(\tilde{X}, \tilde{x}_0)$ . Hence  $|G : H|$  is the number of sheets in the covering.

*Proof.* Let  $[g] \in \pi_1(X, x_0)$ . Let  $\tilde{g}$  be the unique lift of  $g$  starting at  $\tilde{x}_0$ . For  $h \in H$ , its unique lift  $\tilde{h}$  starting at  $\tilde{x}_0$  is a loop. Therefore,  $\tilde{h} \cdot \tilde{g}$  starts at  $\tilde{x}_0$  and ends where  $\tilde{g}$  ends.

Consider  $\Phi : G/H \rightarrow p^{-1}(\{x_0\})$  given by  $\Phi(H[g]) = \tilde{g}(1)$ . Notice that  $p(\tilde{g}(1)) = g(1) = x_0$ .

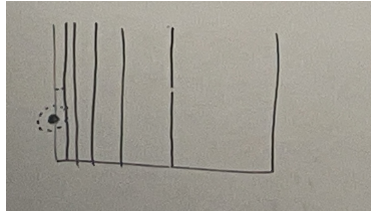
Claim:  $\Phi$  is well-defined.  $H[g_1] = H[g_2] \implies [g_1][g_2]^{-1} = [h] \in H$ . Hence  $[g_1] = [h \cdot g_2] \implies \tilde{g}_1(1) = \tilde{h}\tilde{g}_2(1) = \tilde{g}_2(1)$ . Now let's check the surjectivity of  $\Phi$ . Let  $\tilde{y} \in p^{-1}(\{x_0\}) \implies p(\tilde{y}) = x_0$ . Since  $\tilde{X}$  is path connected, let  $\tilde{g}$  be a path in  $\tilde{X}$  from  $\tilde{x}_0$  to  $\tilde{y}$ . Then,  $g := p \circ \tilde{g}$  is a path in  $X$  from  $p(\tilde{x}_0) = x_0$  to  $p(\tilde{y}) = x_0$ . Hence  $[g] \in G$  and  $\Phi(H[g]) = \tilde{g}(1) = \tilde{y}$ .

It remains to check the injectivity. Consider  $\Phi(H[g_1]) = \Phi(H[g_2])$ . Hence  $\tilde{g}_1(1) = \tilde{g}_2(1) \implies \tilde{g}_1 \cdot \overline{\tilde{g}_2}$  is a loop at  $\tilde{x}_0$ . This means  $[g_1][g_2]^{-1} \in H$ . Thus,  $H[g_1] = H[g_2]$ .  $\square$

**Example 13.15.**  $p : S^1 \rightarrow S^1$ ,  $p(z) = z^2$ , ( $S^1 \subseteq \mathbb{C}$ ). For all  $x \in S^1$ ,  $|p^{-1}(\{x\})| = 2$ , 2-sheeted cover.  $G = \pi_1(S^1, 1)$ ,  $H = p_*\pi_1(S^1, 1)$ ,  $|G : H| = 2$ . Since  $G = \mathbb{Z}$ ,  $H$  is the unique subgroup of index 2,  $2\mathbb{Z}$ .

**Definition 13.16.**  $(X, \mathcal{T})$ . We say  $X$  is locally path-connected iff  $\forall x \in X$ ,  $\forall U \ni x \in \mathcal{T}$ ,  $\exists V \in \mathcal{T}$  such that  $V$  is path-connected and  $x \in V \subseteq U$ .

**Example 13.17.**  $K = \{\frac{1}{n} : n \in \mathbb{N}\}$ . Combine space:  $C = (\{0\} \times [0, 1]) \cup ([0, 1] \times \{0\}) \cup (K \times [0, 1])$ .



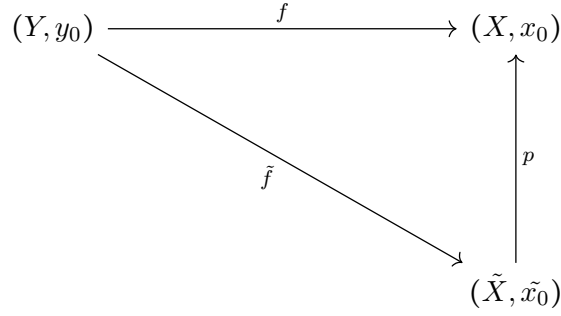
path connected but not locally path connected.

**Proposition 13.18** (Lifting Criterion).  $p : \tilde{X} \rightarrow X$ ,  $p(\tilde{x}_0) = x_0$ .  $Y$  is path-connected and locally path-connected. Let  $f : Y \rightarrow X$  be continuous such that  $f(y_0) = x_0$ . Then a lift

$$\tilde{f} : Y \rightarrow \tilde{X} \text{ with } \tilde{f}(y_0) = \tilde{x}_0 \text{ exists}$$

if and only if

$$f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\tilde{X}, \tilde{x}_0).$$



**Proposition 13.19** (Unique Lifting Property). Suppose  $p : \tilde{X} \rightarrow X$  is a covering space,  $Y$  is connected, and  $f : Y \rightarrow X$  is continuous. Suppose  $\tilde{f}_1, \tilde{f}_2 : Y \rightarrow \tilde{X}$  are lifts of  $f$ . If  $\tilde{f}_1$  and  $\tilde{f}_2$  agree at a point, then  $\tilde{f}_1 = \tilde{f}_2$ .

*Proof.* Let  $y \in Y$ , let  $U \subseteq X$  be a nbhd of  $f(y)$  such that  $p^{-1}(U) = \sqcup V_i, p : V_i \rightarrow U$  hom. Let  $\tilde{U}_1$  and  $\tilde{U}_2$  be the  $V_i$ 's containing  $\tilde{f}_1(y)$  and  $\tilde{f}_2(y)$ , respectively. By continuity of  $\tilde{f}_1$  and  $\tilde{f}_2$ , we may find an open nbhd  $N$  of  $y$  such that  $\tilde{f}_1(N) \subseteq \tilde{U}_1$  and  $\tilde{f}_2(N) \subseteq \tilde{U}_2$ .

Case 1:  $\tilde{f}_1(y) \neq \tilde{f}_2(y)$ .

Suppose there exists  $n \in N$  such that

$$\underbrace{\tilde{f}_1(n)}_{\in \tilde{U}_1} = \underbrace{\tilde{f}_2(n)}_{\in \tilde{U}_2} \implies \tilde{U}_1 = \tilde{U}_2$$

We know  $p : \tilde{U}_1 \rightarrow U$  is a bijection, then

$$p \circ \tilde{f}_1(y) \neq p \circ \tilde{f}_2(y) \implies f(y) \neq f(y), \text{ contradict.}$$

Hence  $\forall n \in N, \tilde{f}_1(n) \neq \tilde{f}_2(n)$ . Thus,  $\{y \in Y : \tilde{f}_1(y) \neq \tilde{f}_2(y)\}$  is open.

Case 2:  $\tilde{f}_1(y) = \tilde{f}_2(y)$ .

Then  $\tilde{U}_1 = \tilde{U}_2$ , hence  $\forall n \in N, \tilde{f}_1(n), \tilde{f}_2(n) \in \tilde{U}_1$ . Since  $p \circ \tilde{f}_1(n) = p \circ \tilde{f}_2(n) \implies \tilde{f}_1(n) = \tilde{f}_2(n)$ .

Thus,  $\{y \in Y : \tilde{f}_1(y) = \tilde{f}_2(y)\}$  is open. Since  $\tilde{f}_1, \tilde{f}_2$  agree at a point,  $y : \tilde{f}_1(y) = \tilde{f}_2(y)$  is open.

If  $\tilde{f}_1 \neq \tilde{f}_2$ , then  $\{y \in Y : \tilde{f}_1(y) = \tilde{f}_2(y)\}$  clopen. Since  $Y$  is connected,  $\{y : \tilde{f}_1(y) = \tilde{f}_2(y)\} = Y$ , contradict.  $\square$

**Definition 13.20.**  $(X, \mathcal{T})$ . We say  $X$  is semilocally simply connected (SSC) iff

$$\forall x \in X, \exists \underbrace{U}_{x \in U} \in \mathcal{T}, i_* \pi_1(U, x) \rightarrow \pi_1(X, x) \text{ is trivial.}$$

**Definition 13.21.** If  $U$  is simply connected, then  $U$  is SSC.

**Construction:**

Let  $X$  be path connected, locally path connected, SSC. Fix  $x_0 \in X$ . Consider:

$$\tilde{X} = \{[\gamma] : \gamma \text{ path in } X \text{ starting at } x_0\}.$$

Let  $\mathcal{U} = \{U \subseteq X \text{ open, path connected with } \pi_1(U) \rightarrow \pi_1(X) \text{ trivial}\}$ .  
For  $U \in \mathcal{U}$  and  $[\gamma] \in \tilde{X}$ , then

$$U_{[\gamma]} = \{[\gamma \cdot \rho] : \rho \text{ is a path in } U \text{ starting at } \gamma(1)\}.$$

Then  $\mathcal{B} = \{U_{[\gamma]} : U \in \mathcal{U}, \rho \text{ as above}\}$  is a basis for a topology on  $\tilde{X}$ .

## 14 Galois Correspondence

**Proposition 14.1.**  $X$  is path connected, locally path connected, SSC. For all  $H \subseteq \pi_1(X, x_0)$ , there exists a covering  $p : X_H \rightarrow X$  such that  $p_*\pi_1(\tilde{X}_H, \tilde{x}_0) = H$ , for some base point  $\tilde{x}_0$  with  $p(\tilde{x}_0) = x_0$ .

*Proof.* Let  $\tilde{X} = \{[\gamma] : \gamma \text{ path starting at } x_0\}$  be the simply connected covering space of  $X$  we constructed. Consider the equivalent relation on  $\tilde{X}$  given by

$$[\gamma_1] \sim [\gamma_2] \iff \gamma_1(1) = \gamma_2(1) \text{ and } [\gamma_1 \overline{\gamma_2}] \in H.$$

Let  $X_H = \tilde{X} / \sim$ . Observed: suppose  $[\gamma_1], [\gamma_2] \in \tilde{X}$  satisfy  $\gamma_1(1) = \gamma_2(1)$ . Let  $\gamma_1 \in U$  be pc, open, and such that  $\pi_1(U) \rightarrow \pi_1(X)$  trivial. For a path  $\rho \in U$  starting at  $\gamma_1(1)$ , we have that

$$[\gamma_1 \cdot \rho] \sim [\gamma_2 \cdot \rho] \iff [\gamma_1 \rho \overline{\gamma_2 \rho}] \in H \iff [\gamma_1 \overline{\gamma_2}] \in H.$$

Therefore,  $[\gamma_1] \sim [\gamma_2]$  iff every point in  $U_{[\gamma_1]}$  is equivalent to the corresponding point in  $U_{[\gamma_2]}$ . This makes

$$p : X_H \rightarrow X, p([\gamma]) = \gamma(1) \text{ (which should really be } p(\overline{[\gamma]}), \text{ but just say it for abbrev)}$$

a covering space.

Let  $\tilde{x}_0 = \underbrace{[x_0]}_{\text{constant}} \in X_H$ . Consider some  $[\gamma] \in \pi_1(X, x_0)$  and let  $\tilde{\gamma}$  be its unique lift to  $X_H$  starting at  $\tilde{x}_0$ . For  $0 \leq t \leq 1$ , let  $\gamma_t$  be the path in  $X$  from  $x_0$  to  $\gamma(t)$  which follows  $\gamma$ . Then  $t \mapsto [\gamma_t]$  is a path in  $X_H$  from  $[x_0]$  to  $[\gamma]$ . Also,  $p([\gamma_t]) = \gamma_t(1) = \gamma(t)$ . Hence this path is  $\tilde{\gamma}$ . Thus,

$$[\gamma_1] \in p_*\pi_1(X_H, \tilde{x}_0) \iff \tilde{\gamma} \text{ ends at } \tilde{x}_0 \iff [\gamma] = \tilde{x}_0 \text{ in } X_H \iff [\gamma] \in H.$$

□

**Remark 14.2.**  $X_H$  is path-connected because  $\tilde{X}$  is path connected.

**Definition 14.3.**  $p_1 : \tilde{X}_1 \rightarrow X, p_2 : \tilde{X}_2 \rightarrow X$ . An isomorphism between  $p_1$  and  $p_2$  is a homeomorphism  $f : \tilde{X}_1 \rightarrow \tilde{X}_2$  such that  $p_1 = p_2 \circ f$ .

**Proposition 14.4.**  $X$  path connected, locally path connected. Let  $p_1 : \tilde{X}_1 \rightarrow X, p_2 : \tilde{X}_2 \rightarrow X$  be path connected covers of  $X$ . Fix  $\tilde{x}_1 \in p_1^{-1}(x_0), \tilde{x}_2 \in p_2^{-1}(x_0)$ . Then  $p_1, p_2$  are isomorphic via an isomorphism  $\tilde{x}_1 \mapsto \tilde{x}_2$  (basepoint preserving) iff

$$p_{1*}\pi_1(\tilde{X}_1, \tilde{x}_1) = p_{2*}\pi_1(\tilde{X}_2, \tilde{x}_2).$$

*Proof.* If  $f$  is such a homeomorphism, then  $p_1 = p_2 \circ f$  and  $p_2 = p_1 \circ f^{-1}$  immediately, the two subgroups are equal.

Suppose the two subgroups are equal. By the lifting criterion, we can make lifts

$$\begin{aligned}\tilde{p}_1 : \tilde{X}_1 &\rightarrow \tilde{X}_2, \tilde{x}_1 \mapsto \tilde{x}_2, \\ \tilde{p}_2 : \tilde{X}_2 &\rightarrow \tilde{X}_1, \tilde{x}_2 \mapsto \tilde{x}_1.\end{aligned}$$

Note  $p_2 \circ \tilde{p}_1 = p_1$  and  $p_1 \circ \tilde{p}_2 = p_1$ . Consider  $\tilde{p}_1 \circ \tilde{p}_2 : \tilde{X}_2 \rightarrow \tilde{X}_2$ . So

$$p_2 \circ \tilde{p}_1 \circ \tilde{p}_2 = p_1 \circ \tilde{p}_2 = p_2 \implies \tilde{p}_1 \circ \tilde{p}_2 \text{ is a lift of } p_2.$$

Similarly,  $\tilde{p}_2 \circ \tilde{p}_1$  is a lift of  $p_1$ . Also,  $1 : \tilde{X}_1 \rightarrow \tilde{X}_1$  is a lift of  $p_1$  and  $1 : \tilde{X}_2 \rightarrow \tilde{X}_2$  is a lift of  $p_2$ . However, we have

$$\tilde{p}_1 \circ \tilde{p}_2(\tilde{x}_2) = p_1(\tilde{x}_1) = \tilde{x}_2 = 1(\tilde{x}_2) \text{ and } \tilde{p}_2 \circ \tilde{p}_1 = 1(\tilde{x}_1)$$

By the uniqueness of lifts,  $\tilde{p}_1 \circ \tilde{p}_2 = 1, \tilde{p}_2 \circ \tilde{p}_1 = 1$  □

**Recall:**  $X$  pc, lpc, ssc.

$$\mathcal{U} = \{U \subseteq X : U \text{ open, pc, } \pi_1(U) \rightarrow \pi_1(X) \text{ trivial}\}$$

$$U_{[\gamma]} = \{[\gamma \cdot \rho] : \rho \text{ path in } U \text{ starting at } \gamma(1)\}$$

$$\tilde{X} = \{[\gamma] : \text{path in } X \text{ starting at } x_0\}$$

Let  $p : \tilde{X} \rightarrow X, p([\gamma]) = \gamma(1)$ .

**Claim 1:** If  $[\gamma_1] \in U_{[\gamma_2]}$ , then  $U_{[\gamma_1]} = U_{[\gamma_2]}$ .

*Claim 1.* Suppose  $[\gamma_1] = [\gamma_2 \cdot \rho]$ , where  $\rho$  is a path in  $U$ .

( $\subseteq$ ) : If  $[\gamma_1 \cdot \rho_1] \in U_{[\gamma_1]}$ , then  $[\gamma_1 \cdot \rho_1] = [\gamma_2 \cdot \rho \cdot \rho_1] \in U_{[\gamma_2]}$ .

( $\supseteq$ ) : If  $[\gamma_2 \cdot \rho_2] \in U_{[\gamma_2]}$ . Then

$$[\gamma_2 \cdot \rho_2] = [\gamma_2 \cdot \rho \cdot \bar{\rho} \cdot \rho_2] = [\gamma_1 \cdot \bar{\rho} \cdot \rho_2] \in U_{[\gamma_1]}$$

□

**Claim 2:**  $\{U_{[\gamma]}\}$  forms a basis for a topology on  $\tilde{X}$ .

*claim 2.* Let  $[\gamma] \in \tilde{X}$ . Since  $X$  is ssc, there exists  $\gamma(1) \in U \in \mathcal{U}$  and so  $[\gamma] \in U_{[\gamma]}$ . Suppose  $[\gamma] \in U_{[\gamma_1]} \cap V_{[\gamma_2]}$ . By Claim 1,  $U_{[\gamma]} = U_{[\gamma_1]}$  and  $V_{[\gamma]} = V_{[\gamma_2]}$ . Let  $W$  be open and pc with  $W \subseteq U \cap V$  and  $\gamma(1) \in W$ . Then

$$[\gamma] \in W_{[\gamma]} \subseteq \underbrace{U_{[\gamma_1]} \cap V_{[\gamma_2]}}_{U_{[\gamma]} \cap V_{[\gamma]}}$$

□

**Claim 3:**  $p : U_{[x]} \rightarrow U$  is a homeomorphism.

*claim 3.* If  $p([\gamma\rho_1]) = p([\gamma\rho_2])$ , then  $\gamma\rho_1(1) = \gamma\rho_2(1) \implies [\gamma\rho_1] = [\gamma\rho_2]$ .

Since  $U$  is pc,  $p$  is surjective. Let  $V \subseteq U$  be open, and  $[\gamma'] \in U_{[\gamma]}$  with endpoint in  $V$ .

**Subclaim:**  $p^{-1}(V) \cap U_{[\gamma]} = V_{[\gamma']}$ .

*subclaim.* Take  $[\tilde{\gamma}] \in p^{-1}(V) \cap U_{[x]}$ . Then

$$\begin{aligned} [\tilde{\gamma}] &= [\gamma\rho_1], \rho_1 \text{ path in } U \text{ which ends in } V \\ &= [\gamma'\rho_2], \rho_2 \text{ path in } V, \text{ by Claim 1} \\ &\in V_{[\gamma']}. \end{aligned}$$

Take  $[\gamma'\rho] \in V_{[\gamma']}$ , then  $[\gamma'\rho] = [\gamma\rho'\rho] \in U_{[x]}$ , where  $\rho'$  is path in  $U$ . □

Hence  $p([\gamma'\rho]) \in V$ . HW: show that if  $V_{[\gamma']} \subseteq U_{[\gamma]}$ , then  $p(V_{[\gamma']}) = V$  □

**Claim 4:**  $p : \tilde{X} \rightarrow X$  is a covering space.

*Claim 4.* From Claim 3,  $p$  is cts. For  $U \in \mathcal{U}$ ,  $p^{-1}(U) = \bigsqcup_{[\gamma]} U_{[\gamma]}$ . □

**Claim 5:**  $\tilde{X}$  is simply connected.

*Claim 5.* Let  $[\gamma] \in \tilde{X}$  and let  $\gamma_t$  be the path in  $X$  from  $x_0$  to  $\gamma(t)$ . Then,  $t \mapsto [\gamma_t]$  is a lift of  $\gamma$  from  $[x_0]$  to  $[\gamma]$ . Thus,  $\tilde{X}$  is pc.

Since  $p_*$  is injective, it is enough to show that  $p_*\pi_1(\tilde{X}, [x_0]) = \{[x_0]\}$ . Take  $[\gamma] \in p_*\pi_1(\tilde{X}, [x_0])$ . Therefore its lift to  $\tilde{X}$  (before) is a loop, i.e.,  $[x_0] = [\gamma]$ , which completes the proof. □

**Remark 14.5.** By the Galois correspondence,  $\tilde{X}$  is the unique simply connected covering space of  $X$ , up to isomorphism. We call it the universal cover of  $X$ .